

On the Harborth Conjecture

Part I

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Abstract

Harborth's conjecture states that every planar graph has a crossing-free straight-line drawing in which every edge has an integer length. Kleber's strengthening asks for the vertices themselves to have integer coordinates. In this series of papers, we make progress towards settling these conjectures. We reduce Kleber's conjecture to local rational-distance statements for special polygons with at most five vertices. The triangle case is known from the results of Almering and Berry. In this paper, we prove the existence of a rational-distance point on an interior integer diagonal in all the cases for non-degenerate quadrilaterals. In the upcoming papers, we focus on non-degenerate pentagons and then degenerate quadrilaterals and degenerate pentagons.

Keywords: algebraic number theory, algebraic geometry, planar graphs, integral drawings, elliptic curves, rational distances, rational points.

1 Introduction

Harborth's conjecture [1] states that every planar graph has a crossing-free straight-line drawing in which every edge has integer length. Kleber's strengthening asks for the vertices themselves to lie on the integer lattice. In this paper, we call the latter an *integral drawing*: all vertices lie in $\mathbb{Z}^2$, all edges are straight-line segments, no two edges cross except at common endpoints, and every edge has integer length.

The *Kemnitz-Harborth algorithm* is inductive. It is enough to consider triangulated planar graphs. Every triangulation has a vertex of degree at most five. Delete such a vertex v , draw the smaller graph, and then put v back inside the face bounded by its neighbors. Since v has degree $d \leq 5$, the local face is a triangle, quadrilateral, or pentagon.

The graph-theoretic part of the method is elementary. The hard part is Diophantine geometry. We need a point inside the relevant local polygon whose distances to all boundary vertices are rational. If such a point has rational coordinates, then after a final global scaling all coordinates and all edge lengths become integers.

1.1 Basics

Definition 1.1 (Plane straight-line drawing). Let $G = (V, E)$ be a finite simple graph. A straight-line drawing of G is an injective map

$$\varphi : V \longrightarrow \mathbb{R}^2$$

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in which each edge $uv \in E$ is represented by the segment $[\varphi(u), \varphi(v)]$. The drawing is plane if two edge segments meet only when the corresponding graph edges share an endpoint, and only at that endpoint.

Definition 1.2 (Integral drawing). An integral drawing of G is a plane straight-line drawing $\varphi : V(G) \rightarrow \mathbb{Z}^2$ such that

$$\|\varphi(u) - \varphi(v)\| \in \mathbb{Z} \quad \text{for every } uv \in E(G).$$

Definition 1.3 (Rational drawing). A rational drawing of G is a plane straight-line drawing $\varphi : V(G) \rightarrow \mathbb{Q}^2$ such that

$$\|\varphi(u) - \varphi(v)\| \in \mathbb{Q} \quad \text{for every } uv \in E(G).$$

Rational drawings are used as an intermediate object. A single scaling turns them into an integral drawing.

Lemma 1.4 (Scaling). *If G has a rational drawing, then G has an integral drawing.*

Proof. Choose a positive integer M divisible by the denominators of all vertex coordinates and by the denominators of all edge lengths. Replacing every point p by Mp sends vertices to $\mathbb{Z}^2$. It also multiplies every edge length by M , so all edge lengths become integers. Scaling preserves straightness and crossing-freeness. $\square$

Conjecture 1.5 (Harborth [1]). *Every planar graph has a plane straight-line drawing whose edge lengths are integers.*

Conjecture 1.6 (Kleber [13]). *Every planar graph has an integral drawing.*

1.2 Planar graphs

Definition 1.7 (Triangulation). A plane graph is a simple planar graph together with a fixed crossing-free topological embedding. A triangulation is a plane graph in which every face, including the outer face, is bounded by a triangle.

Lemma 1.8 (Triangulating a plane graph). *Every simple plane graph with at least three vertices can be extended, by adding only non-crossing edges and no new vertices, to a plane triangulation.*

This standard lemma justifies reducing drawing questions for planar graphs to triangulations: once a drawing has been found for the triangulated supergraph, the added edges may be deleted.

Lemma 1.9 (Euler degree bound). *Every non-empty simple planar graph has a vertex of degree at most five.*

Lemma 1.10 (The polygon around a deleted vertex). *Let T be a plane triangulation and let v be a vertex of degree d . List the neighbors of v in cyclic order around v as*

$$v_1, v_2, \dots, v_d.$$

Then $v_1v_2 \cdots v_dv_1$ is a cycle in $T - v$, and this cycle bounds the face created by deleting v .

Proof. In a triangulation, each face incident with v is a triangle

$$vv_iv_{i+1},$$

with indices modulo d . Thus v_i is adjacent to v_{i+1} for each i . Removing v merges the d triangular faces incident with v into one face whose boundary is the cycle

$$v_1v_2 \cdots v_dv_1.$$

□

1.3 Rational-distance points

A line is called rational if it is defined by a linear equation with rational coefficients. A line segment has rational length if its Euclidean length lies in $\mathbb{Q}$.

Definition 1.11 (Rational point and rational-distance point). A point $p = (x, y) \in \mathbb{R}^2$ is rational if $x, y \in \mathbb{Q}$. Let $S = \{p_1, \dots, p_d\} \subset \mathbb{R}^2$. A point $q \in \mathbb{R}^2$ is a rational-distance point for S if

$$\|q - p_i\| \in \mathbb{Q} \quad \text{for every } i = 1, \dots, d.$$

Lemma 1.12 (Three centers force rational coordinates). *Let $A, B, C \in \mathbb{Q}^2$ be non-collinear. If $p \in \mathbb{R}^2$ satisfies*

$$\|p - A\|, \quad \|p - B\|, \quad \|p - C\| \in \mathbb{Q},$$

then $p \in \mathbb{Q}^2$. In other words, if the three distances from p to A, B, C are rational, then p has rational coordinates.

Lemma 1.13 (Integer and rational formulations). *Suppose a local insertion theorem is proved for every polygon with integer coordinates and integer side lengths. Then the corresponding theorem also holds for every polygon with rational coordinates and rational side lengths, provided the statement is invariant under scaling.*

1.4 Polygons

Let $n \geq 3$, and let $P = v_1v_2 \cdots v_nv_1$ be a closed polygonal chain in $\mathbb{R}^2$. Indices are taken cyclically modulo n , so that $v_{n+1} = v_1$. The sides of P are the closed line segments $e_i = [v_i, v_{i+1}]$, $i = 1, \dots, n$. Two sides e_i and e_j are called adjacent if they share an endpoint in the cyclic order, that is, if $j = i + 1$, $i = j + 1$, or $\{i, j\} = \{1, n\}$. Otherwise they are called non-adjacent.

Definition 1.14 (Simple polygon). The polygonal chain $P = v_1v_2 \cdots v_nv_1$ is a *simple polygon* if its boundary has no self-intersections. Equivalently: the vertices $v_1, \dots, v_n$ are distinct; every side has positive length, i.e. $v_i \neq v_{i+1}$ for every i ; two adjacent sides intersect only in their common endpoint; two non-adjacent sides are disjoint. Thus a simple polygon may be convex or concave, but it has no repeated vertices, no crossing sides, no overlapping sides, and no vertex lying in the relative interior of a non-incident side.

For a simple polygon P , we also write P for the compact polygonal region consisting of its boundary together with its bounded interior.

Definition 1.15 (Non-degenerate polygon). A polygon $P = v_1v_2 \cdots v_nv_1$ is *non-degenerate* if it is simple and has no flat boundary angle. Equivalently, P is simple and no three consecutive vertices are collinear: v_{i-1}, v_i, v_{i+1} are not collinear for every i . In this convention, every interior angle of P is strictly between 0 and 2π , and no interior angle is equal to π .

1.5 Triangles

Theorem 1.16 (Almering [10], Berry [11]). *Let $A, B, C \in \mathbb{R}^2$ be non-collinear. Suppose that*

$$\|A - B\| \in \mathbb{Q}, \quad \|A - C\|^2 \in \mathbb{Q}, \quad \|B - C\|^2 \in \mathbb{Q}.$$

Then the points whose distances from A, B, C are all rational are dense in $\mathbb{R}^2$.

Almering [10] proved the classical rational-triangle case. Berry [11] proved the version stated above. In this paper, we only need the following consequence: every triangle with rational coordinates and rational side lengths contains an interior rational point whose distances to the three vertices are rational.

Corollary 1.17 (Triangle insertion). *Let ΔABC be a non-degenerate triangle with*

$$A, B, C \in \mathbb{Q}^2, \quad \|A - B\|, \|B - C\|, \|C - A\| \in \mathbb{Q}.$$

Then the interior of ΔABC contains a rational point $p \in \mathbb{Q}^2$ such that

$$\|p - A\|, \quad \|p - B\|, \quad \|p - C\| \in \mathbb{Q}.$$

1.6 Quadrilaterals

A quadrilateral is an ordered four-cycle with straight sides. It is *simple* if its boundary is a simple closed polygonal curve. A simple quadrilateral is *convex* if the union of its boundary and bounded interior is a convex subset of $\mathbb{R}^2$. It is *concave* if it is simple but not convex; equivalently, one of its interior angles is reflex. A simple quadrilateral is called non-degenerate if no three of its vertices are collinear.

Definition 1.18 (Interior diagonal). Let Q be a simple quadrilateral. A diagonal is a segment joining two non-adjacent vertices. A diagonal is an interior diagonal if its relative interior is contained in the interior of Q .

Definition 1.19 (Diagonal rational-distance sets). Let $U, V, A, B \in \mathbb{Q}^2$, and suppose that UV is the chosen diagonal. We write

$$\mathcal{R}^\circ(U, V; A, B)$$

for the set of rational points P on the open segment UV such that

$$PU, \quad PV, \quad PA, \quad PB$$

are all rational. We write $\mathcal{R}(U, V; A, B)$ for the analogous set on the closed segment UV .

Lemma 1.20 (An interior diagonal sees the quadrilateral). *Let $Q = ABCD$ be a simple quadrilateral, and suppose that AC is an interior diagonal. If p lies on the open segment AC , then the segments from p to A, B, C, D are contained in Q . Thus placing a new vertex at p and joining it to all four vertices creates no crossings inside Q .*

Proof. The diagonal AC splits the simple quadrilateral into the two triangles ΔABC and ΔACD . If $p \in AC$, then

$$[p, A] \cup [p, B] \cup [p, C] \subset \Delta ABC$$

and

$$[p, A] \cup [p, C] \cup [p, D] \subset \Delta ACD.$$

Both triangles lie in Q , so all four segments from p to the vertices of Q lie in Q . $\square$

Conjecture 1.21 (Quadrilateral diagonal insertion). *Let $Q = ABCD$ be a simple quadrilateral with*

$$A, B, C, D \in \mathbb{Z}^2, \quad AB, BC, CD, DA \in \mathbb{Z},$$

and suppose that Q has an interior diagonal of integer length. Then the open segment of that diagonal contains a rational point $x \in \mathbb{Q}^2$ such that

$$\|x - A\|, \quad \|x - B\|, \quad \|x - C\|, \quad \|x - D\| \in \mathbb{Q}.$$

By Lemma 1.13, the same conjecture may equivalently be used in rational-coordinate, rational-length form. This is the form needed inside the induction, because the intermediate drawing before the final scaling is rational rather than necessarily integral.

1.7 Pentagons

For pentagons, being merely on the interior diagonal or inside the polygon is not enough. The new vertex must see all five boundary vertices. This is a place where the kernel is needed.

Definition 1.22 (Kernel of a pentagon). Let $P = A_1A_2A_3A_4A_5$ be a simple pentagon. The kernel of P , denoted $\ker(P)$, is the set of points $q \in P$ such that each segment

$$[q, A_i] \quad (i = 1, \dots, 5)$$

is contained in P .

Theorem 1.23 (Pentagons are star-shaped). *Every simple pentagon has non-empty kernel.*

Proof. This is the $n = 5$ case of the art gallery theorem of Chvatal: every simple polygon with n vertices can be guarded by $\lfloor n/3 \rfloor$ points [12]. For $n = 5$, one point is enough. A single guard point is precisely a point from which the whole polygon is visible, hence a point of the kernel. $\square$

Conjecture 1.24 (Pentagon kernel insertion). *Let $P = A_1A_2A_3A_4A_5$ be a simple pentagon with*

$$A_i \in \mathbb{Z}^2 \quad (i = 1, \dots, 5), \quad A_iA_{i+1} \in \mathbb{Z} \quad (i = 1, \dots, 5),$$

where indices are taken modulo 5. Then $\ker(P)$ contains a rational point $x \in \mathbb{Q}^2$ such that

$$\|x - A_i\| \in \mathbb{Q} \quad (i = 1, \dots, 5).$$

(Diagonal version) A useful sufficient variant is the following: If one can prove that every integer-coordinate, integer-sided pentagon contains such a point on an interior diagonal segment contained in $\ker(P)$, then Conjecture 1.24 follows for the purposes of the induction. The condition that the segment lie in the kernel is essential: a point on an arbitrary interior diagonal need not see all five vertices.

As in the quadrilateral case, Lemma 1.13 converts the integer statement into the rational statement needed during the induction.

1.8 Kemnitz–Harborth induction

Theorem 1.25. *Assume Conjecture 1.21 and Conjecture 1.24. Then every planar graph has an integral drawing.*

Proof. Graphs with at most two vertices have integral drawings immediately. For graphs with at least three vertices, Lemma 1.8 shows that it is enough to prove the result for plane triangulations, because deleting added edges from an integral drawing preserves an integral drawing of the original graph.

We prove, by induction on the number of vertices, that every plane triangulation has a rational drawing preserving the given embedding. The base case is a single triangular face, which follows from Corollary 1.17. At the end, Lemma 1.4 converts the rational drawing into an integral drawing.

Let T be a plane triangulation with at least four vertices. By Lemma 1.9, T has a vertex v of degree $d \leq 5$. In a triangulation with at least four vertices, $d \geq 3$. Let

$$v_1, \dots, v_d$$

be the cyclic order of the neighbors of v . By Lemma 1.10, deleting v creates a face bounded by the cycle

$$C = v_1 v_2 \cdots v_d v_1.$$

If $d = 3$, the graph $T - v$ is already a plane triangulation. By induction it has a rational drawing. The cycle C is drawn as a triangle with rational vertex coordinates and rational side lengths. By Corollary 1.17, there is a rational point inside this triangle at rational distance from its three vertices. Placing v at this point and joining it to v_1, v_2, v_3 creates no crossings and preserves rational edge lengths.

If $d = 4$, add one auxiliary diagonal inside the quadrilateral face bounded by C , obtaining a plane triangulation T' on the vertex set $V(T) \setminus \{v\}$. By induction, T' has a rational drawing. The boundary cycle C is drawn as a simple quadrilateral with rational side lengths, and the added diagonal is drawn as an interior diagonal of rational length. By the rational form of Conjecture 1.21, that diagonal contains a rational point at rational distance from all four boundary vertices. By Lemma 1.20, placing v at this point and joining it to the four boundary vertices creates no crossings. Deleting the auxiliary diagonal gives a rational drawing of T .

If $d = 5$, add two non-crossing auxiliary diagonals inside the pentagonal face bounded by C , obtaining a plane triangulation T' on $V(T) \setminus \{v\}$. By induction, T' has a rational drawing. The boundary cycle C is drawn as a simple pentagon with rational side lengths. By the rational form of Conjecture 1.24, its kernel contains a rational point at rational distance from all five boundary vertices. Placing v at this point and joining it to $v_1, \dots, v_5$ creates no crossings by the definition of the kernel. Deleting the two auxiliary diagonals gives a rational drawing of T .

This completes the induction. Finally, apply Lemma 1.4. □

1.9 Main theorem

Theorem 1.26 (Main quadrilateral theorem). *Let Q be a simple **non-degenerate** quadrilateral, convex or concave, in $\mathbb{R}^2$. Assume that the four vertices of Q have integer coordinates, the four side lengths of Q are integers, and an interior diagonal UV has integer length. Then there exists a rational point P on the open segment UV such that the four distances from P to the vertices of Q are rational. In particular, P lies in the interior of Q .*

The theorem is independent of orientation. A rational Euclidean normalization sends the chosen diagonal to the x -axis and preserves both rationality of coordinates and Euclidean distances. After this normalization, the problem becomes a rational-distance problem on a line, and our main result is the endpoint-line theorem stated below.

The proof is organized as follows. Sections 2 and 3 reduce the problem to a rational-distance problem on the x -axis and state the endpoint line theorem. Section 4 proves the non-exceptional line case from Love's detour theorem. Sections 5 and 6 handle the convex exceptional cases. Section 7 handles the remaining exceptional endpoint-line cases and completes the proof of the endpoint line theorem. Section 8 then applies the line theorem and the convex exceptional arguments to prove Theorem 1.26. Within the relevant case sections we also prove stronger infinitude theorems and provide two finite rank-zero examples. Readers interested only in the existence theorem may skip those sub-sections on a first reading. The appendix provides the Case III verification details.

2 Normalization

Lemma 2.1 (Rational normalization). *Let $U, V \in \mathbb{Q}^2$, and suppose that*

$$D = \|V - U\|$$

is a positive rational number. Then there is an affine Euclidean isometry

$$F : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

whose matrix and translation vector have rational entries such that

$$F(U) = (0, 0), \quad F(V) = (D, 0).$$

The inverse map F^{-1} also has rational coefficients. Consequently F carries $\mathbb{Q}^2$ bijectively onto $\mathbb{Q}^2$ and preserves all Euclidean distances.

Proof. Translate U to the origin. Write

$$V - U = (m, n), \quad m, n \in \mathbb{Q}.$$

Since $D^2 = m^2 + n^2$, the matrix

$$R = \frac{1}{D} \begin{pmatrix} m & n \\ -n & m \end{pmatrix}$$

has rational entries. It is orthogonal because

$$R^T R = \frac{1}{D^2} \begin{pmatrix} m & -n \\ n & m \end{pmatrix} \begin{pmatrix} m & n \\ -n & m \end{pmatrix} \frac{m^2 + n^2}{D^2} II.$$

Also

$$R(V - U) = (D, 0).$$

Thus $F(X) = R(X - U)$ has the desired properties. Since $R^{-1} = R^T$, the inverse affine map is again rational. $\square$

Definition 2.2 (The set $\mathcal{S}$). Set

$$\mathcal{S} = \{\alpha \in \mathbb{Q} : 1 + \alpha^2 \text{ is a square in } \mathbb{Q}\}.$$

Lemma 2.3 (The elementary distance criterion). *Let $C = (e, f) \in \mathbb{Q}^2$ with $f \neq 0$, and let $P = (x, 0)$ with $x \in \mathbb{Q}$. Then $PC \in \mathbb{Q}$ if and only if*

$$\frac{x - e}{f} \in \mathcal{S}.$$

Proof. Put $\alpha = (x - e)/f$. Then

$$PC^2 = (x - e)^2 + f^2 = f^2(1 + \alpha^2).$$

Since $f \in \mathbb{Q}^\times$, the distance PC is rational if and only if the nonnegative number $1 + \alpha^2$ is the square of a rational number. $\square$

Lemma 2.4 (Parametrization of $\mathcal{S}$). *One has*

$$\mathcal{S} = \left\{ \frac{t - t^{-1}}{2} : t \in \mathbb{Q}^\times \right\}.$$

In particular, $\mathcal{S}$ is dense in $\mathbb{R}$. Moreover, $\mathcal{S} = -\mathcal{S}$.

Proof. If $\alpha = (t - t^{-1})/2$, then

$$1 + \alpha^2 = \left(\frac{t + t^{-1}}{2} \right)^2.$$

Conversely, suppose $1 + \alpha^2 = h^2$ with $h \in \mathbb{Q}$. Then

$$(h + \alpha)(h - \alpha) = 1.$$

Taking $t = h + \alpha$ gives $t^{-1} = h - \alpha$, and therefore

$$\alpha = \frac{t - t^{-1}}{2}.$$

The map

$$\phi : (0, \infty) \longrightarrow \mathbb{R}, \quad \phi(t) = \frac{t - t^{-1}}{2}$$

is continuous and strictly increasing, because

$$\phi'(t) = \frac{1 + t^{-2}}{2} > 0.$$

Moreover

$$\lim_{t \rightarrow 0^+} \phi(t) = -\infty, \quad \lim_{t \rightarrow \infty} \phi(t) = +\infty.$$

Thus ϕ maps $(0, \infty)$ homeomorphically onto $\mathbb{R}$. Since $\mathbb{Q}_{>0}$ is dense in $(0, \infty)$, its image $\phi(\mathbb{Q}_{>0}) \subset \mathcal{S}$ is dense in $\mathbb{R}$. Hence $\mathcal{S}$ is dense in $\mathbb{R}$. Finally, the defining condition for $\mathcal{S}$ is unchanged when α is replaced by $-\alpha$, so $\mathcal{S} = -\mathcal{S}$. $\square$

3 The endpoint line theorem

Theorem 3.1 (Endpoint line theorem). *Let U, V be distinct rational points on a rational line L , and let*

$$A, B \in \mathbb{Q}^2 \setminus L$$

lie on opposite sides of L . Suppose that

$$UV, UA, VA, UB, VB \in \mathbb{Q}.$$

Assume also that neither U nor V lies on the line AB . Then the open segment UV contains a rational point P such that

$$PA, PB \in \mathbb{Q}.$$

Consequently PU, PV, PA, PB are all rational.

After applying Lemma 2.1, and then, if necessary, reflecting in the x -axis and interchanging the names of A and B , it is enough to prove Theorem 3.1 in the following normalized form, where L is the x -axis:

$$\begin{aligned} U &= (0, 0), & V &= (D, 0), & D &\in \mathbb{Q}_{>0}, \\ A &= (a, b), & B &= (c, d), & a, b, c, d &\in \mathbb{Q}, & b > 0 > d. \end{aligned}$$

The additional nondegeneracy hypothesis is that neither endpoint $(0, 0)$ nor $(D, 0)$ lies on the line through A and B . For $P = (x, 0)$, define

$$\alpha_1 = \frac{x - a}{b}, \quad \alpha_2 = \frac{c - x}{d},$$

and set

$$s = \frac{b}{d}, \quad r = \frac{c - a}{d}.$$

Then $s < 0$.

Lemma 3.2 (The detour equation). *With the notation above,*

$$s\alpha_1 + \alpha_2 = r. \tag{3.1}$$

Moreover, $P = (x, 0)$ has rational distance from both A and B if and only if

$$\alpha_1, \alpha_2 \in \mathcal{S}.$$

Finally, P lies on the open segment UV if and only if

$$\alpha_1 \in I := \left(-\frac{a}{b}, \frac{D - a}{b} \right).$$

Proof. The linear relation is immediate:

$$s\alpha_1 + \alpha_2 = \frac{b}{d} \frac{x - a}{b} + \frac{c - x}{d} \frac{c - a}{d} r.$$

The rational-distance assertion is Lemma 2.3, applied first to A and then to B . For B , Lemma 2.3 uses $(x - c)/d$, while our α_2 is its negative; this is equivalent because $\mathcal{S}$ is closed under negation. Finally, since $x = a + b\alpha_1$ and $b > 0$, the inequality $0 < x < D$ is equivalent to

$$-\frac{a}{b} < \alpha_1 < \frac{D - a}{b}.$$

□

The endpoint hypotheses say exactly that the two endpoint values

$$\lambda = -\frac{a}{b}, \quad \mu = \frac{D - a}{b}$$

are rational detour solutions:

$$\lambda, \mu \in \mathcal{S}, \quad r - s\lambda, r - s\mu \in \mathcal{S}.$$

Indeed, these are the values of α_1 at $P = U$ and $P = V$, and the rationality of UA, VA, UB, VB is translated by Lemma 2.3. As in Lemma 3.2, the sign convention in the B -coordinate does not change the condition, since $\mathcal{S} = -\mathcal{S}$. The endpoint line theorem asks for a rational detour solution with α_1 strictly between λ and μ . Its proof is completed in three stages below: the non-exceptional case, the convex exceptional cases needed later in the quadrilateral argument, and the remaining exceptional endpoint-line cases.

4 The generic endpoint-line case

We use the following form of Love's detour theorem.

Theorem 4.1 (Love's detour theorem[3]). *Let $r, s \in \mathbb{Q}$, $s \neq 0$, and consider*

$$s\alpha_1 + \alpha_2 = r, \quad \alpha_1, \alpha_2 \in \mathcal{S}.$$

Associated to this equation is the elliptic curve

$$E_{r,s}: \quad y^2 = x^3 + (1 + r^2 + s^2)x^2 + s^2x$$

and the rational point

$$R = (-1, r).$$

If

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1 - s^2)^2,$$

then R is non-torsion. Hence rational detour solutions are dense on the corresponding real detour component, for each fixed choice of signs of the two square roots.

Proof. This is the detour case of Love's theorem on rational configuration problems; see [3, Theorem 1.4 and Section 5]. The density statement is the standard consequence that the subgroup generated by a non-torsion point is dense in the connected component of the identity in the real Lie group $E_{r,s}(\mathbb{R})$, and hence dense in the real component containing that point after translation. Passing back through the detour parametrization preserves density on the corresponding real sign branch. □

Proposition 4.2 (Generic endpoint line theorem). *Assume the normalized endpoint-line setup. If*

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1 - s^2)^2,$$

then there are infinitely many rational points P on the open segment UV such that $PA, PB \in \mathbb{Q}$.

Proof. Let

$$I = \left(-\frac{a}{b}, \frac{D-a}{b} \right).$$

The endpoint hypotheses give rational detour points at the two endpoints of the nonempty interval I . At those endpoints, choose the positive rational values of the two square roots

$$\sqrt{1 + \alpha_1^2}, \quad \sqrt{1 + (r - s\alpha_1)^2}$$

and then take the real branch on which these square roots remain positive. This branch is parameterized continuously by the real variable α_1 and has no singularity over a finite value of α_1 , so the endpoint points lie on the same real branch. By Theorem 4.1, rational detour solutions are dense on this branch. Since I is open and nonempty, there are infinitely many rational detour solutions with $\alpha_1 \in I$. For each such solution, put

$$x = a + b\alpha_1, \quad P = (x, 0).$$

Lemma 3.2 gives $0 < x < D$ and $PA, PB \in \mathbb{Q}$. □

5 Elementary exceptional cases for convex quadrilaterals

In this section we give the elementary convex proofs of the exceptional configurations. We assume the normalized setup

$$\begin{aligned} U &= (0, 0), & V &= (D, 0), & D &\in \mathbb{Q}_{>0}, \\ A &= (a, b), & B &= (c, d), & b &> 0 > d, \end{aligned}$$

with

$$s = \frac{b}{d}, \quad r = \frac{c-a}{d}.$$

We assume in addition that the quadrilateral

$$Q = UAVB$$

with diagonal UV is simple, non-degenerate, and convex. Hence the other diagonal AB meets UV in the relative interior of both diagonals.

Since $s < 0$, the exceptional conditions in Theorem 4.1 reduce to

$$r = 0, \quad s = -1, \quad 4r^2s = -(1 - s^2)^2. \tag{5.1}$$

Indeed, the equality

$$4r^2s = (1 - s^2)^2$$

has nonpositive left side and nonnegative right side. Therefore it can hold only when both sides are zero, i.e. when $r = 0$ and $s = -1$, already included in the first two cases in (5.1).

Proposition 5.1 (The case $r = 0$). *Assume the normalized convex quadrilateral setup. If*

$$r = 0,$$

then the open segment UV contains a rational point whose distances from U, V, A, B are rational.

Proof. The condition $r = 0$ means

$$c = a.$$

Thus the second diagonal AB is vertical and intersects the x -axis at

$$X = (a, 0).$$

In a convex quadrilateral, the two diagonals intersect in their relative interiors. Hence X lies on the open segment UV , so

$$0 < a < D.$$

The distances from X to U and V are

$$XU = a, \quad XV = D - a,$$

which are rational. The distances from X to A and B are

$$XA = |b| = b, \quad XB = |d| = -d,$$

which are rational because $b, d \in \mathbb{Q}$. Therefore X is a required point. $\square$

5.1 A finite aligned $r = 0$ example

Example 5.2. Let

$$U = (0, 0), \quad A = (192, 144), \quad V = (384, 0), \quad B = (192, -256).$$

Then $UAVB$ is a convex kite whose chosen diagonal is UV . The side lengths are

$$UA = AV = 240, \quad VB = BU = 320,$$

and

$$UV = 384.$$

Thus this is an integer-coordinate quadrilateral with integer side lengths and an integer interior diagonal.

Theorem 5.3 (A rank-zero finite aligned example). *For the quadrilateral in Example 5.2,*

$$\mathcal{R}^\circ(U, V; A, B) = \{(192, 0)\}.$$

Proof. Here the normalized data are

$$D = 384, \quad a = c = 192, \quad b = 144, \quad d = -256.$$

Therefore

$$r = \frac{c - a}{d} = 0, \quad s = \frac{b}{d} = -\frac{9}{16}, \quad \rho = -s = \frac{9}{16}.$$

For $P = (x, 0)$, put

$$\alpha = \frac{x - 192}{144}.$$

Then $P \in UV^\circ$ if and only if

$$-\frac{4}{3} < \alpha < \frac{4}{3}.$$

Moreover

$$PA^2 = 144^2(1 + \alpha^2),$$

and, since $B = (192, -256)$,

$$PB^2 = 256^2 \left(1 + \left(\frac{144}{256} \alpha \right)^2 \right) 256^2(1 + \rho^2 \alpha^2).$$

Thus PA and PB are rational if and only if

$$1 + \alpha^2, \quad 1 + \rho^2 \alpha^2$$

are rational squares.

The associated concordant-form elliptic curve is

$$E_\rho : \quad \eta^2 = \xi(\xi + \rho^2)(\xi + 1)\xi \left(\xi + \frac{81}{256} \right) (\xi + 1).$$

With

$$X = 256\xi, \quad Y = 4096\eta,$$

this becomes

$$E : \quad Y^2 = X(X + 81)(X + 256).$$

We use the Cremona-LMFDB rank-zero [2, 6]:

$$\text{rank } E(\mathbb{Q}) = 0, \quad E(\mathbb{Q}) * \text{tors} \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}.$$

Hence every rational point of $E * \rho$, equivalently of E , is torsion.

The torsion-value calculation for the aligned $r = 0$ concordant-form case, as in Selmer–Spindler’s description of concordant forms [8, Theorem 2 and Theorem 4], says that the finite α -values coming from rational torsion are

$$\alpha = 0 \quad \text{and, since } \rho = \left(\frac{3}{4} \right)^2, \quad \alpha = \pm \frac{1}{\sqrt{\rho}} \pm \frac{4}{3}.$$

Because the curve has rank 0, there are no non-torsion rational points and hence no other rational α -values.

The values $\alpha = \pm 4/3$ are precisely the two endpoints of the interval:

$$\alpha = -\frac{4}{3} \iff x = 192 + 144 \left(-\frac{4}{3} \right) = 0,$$

and

$$\alpha = \frac{4}{3} \iff x = 192 + 144 \left(\frac{4}{3} \right) = 384.$$

The only torsion value strictly inside the interval is $\alpha = 0$, giving

$$x = 192.$$

Thus the only interior point is

$$P = (192, 0).$$

It indeed works, since

$$PU = PV = 192, \quad PA = 144, \quad PB = 256.$$

Therefore

$$\mathcal{R}^\circ(U, V; A, B) = \{(192, 0)\}.$$

□

Remark 5.4 (Convex $r = 0$ can be finite). Theorem 5.3 gives an explicit affirmative answer to the question whether the convex $r = 0$ case can have only finitely many rational points on the interior diagonal. In fact, the example has exactly one such interior point. Thus the convex $r = 0$ branch of the existence proof in Proposition 5.1 should not be interpreted as always producing infinitely many points. Additional arithmetic information about the associated concordant-form elliptic curve is needed. Positive-rank specializations can yield infinitely many points on suitable real intervals, whereas in the rank-zero specialization above the torsion table leaves only the two endpoints and the single interior point $(192, 0)$.

Proposition 5.5 (The exceptional detour identity). *Assume the normalized convex quadrilateral setup. Suppose*

$$r \neq 0, \quad s \neq -1, \quad 4r^2s = -(1 - s^2)^2. \quad (5.2)$$

Then the open segment UV contains a rational point whose distances from U, V, A, B are rational.

Proof. Let $X = (x, 0)$ be the intersection point of the diagonals AB and UV . Since Q is convex, X lies on the open segment UV .

The line through $A = (a, b)$ and $B = (c, d)$ meets the x -axis at X . Using similar triangles, or equivalently the equation of the line, we get

$$\frac{x - a}{b} = \frac{c - a}{b - d}.$$

Thus

$$\alpha_1 := \frac{x - a}{b} \frac{c - a}{b - d}.$$

Since $s = b/d$ and $r = (c - a)/d$, this becomes

$$\alpha_1 = \frac{r}{s - 1}. \quad (5.3)$$

At the same intersection point, the parameters along the line AB give

$$\alpha_2 := \frac{c - x}{d} = -\alpha_1. \quad (5.4)$$

Define

$$t = \frac{1 - s^2}{2r}.$$

Because $r \neq 0$ and $s \neq -1$, and because $s < 0$, we have $t \neq 0$. From (5.2),

$$t^2 = \frac{(1 - s^2)^2}{4r^2} s.$$

Therefore

$$\frac{t - t^{-1}}{2} \frac{t^2 - 1}{2t} \frac{-s - 1}{(1 - s^2)/r} \frac{r}{s - 1} \alpha_1.$$

By Lemma 2.4, $\alpha_1 \in \mathcal{S}$. Since $\mathcal{S}$ is closed under negation, (5.4) gives $\alpha_2 \in \mathcal{S}$. Therefore XA and XB are rational by Lemma 3.2. The point X has rational coordinates by (5.3), so $XU = x$ and $XV = D - x$ are rational. Thus X is a required point. $\square$

5.2 Infinitude in the convex quadratic-exceptional case

Proposition 5.5 uses only the diagonal-intersection point, but the Case III torsion table upgrades the conclusion to infinitely many points.

Proposition 5.6 (Convex quadratic exceptional identity gives infinitely many). *Assume the normalized convex quadrilateral setup*

$$\begin{aligned} U &= (0, 0), & V &= (D, 0), \\ A &= (a, b), & B &= (c, d), & b &> 0 > d, \end{aligned}$$

with

$$s = \frac{b}{d}, \quad r = \frac{c - a}{d}.$$

Suppose

$$r \neq 0, \quad s \neq -1, \quad 4r^2 s = -(1 - s^2)^2.$$

Assume also that the side lengths

$$UA, \quad AV, \quad VB, \quad BU$$

are rational. Then the open diagonal segment UV° contains infinitely many rational points whose distances from U, V, A, B are rational.

Proof. Let

$$\alpha = \frac{x - a}{b}$$

for a point $P = (x, 0)$ on the diagonal line. As in the endpoint-line setup, the rational-distance condition from A and B is encoded by the detour equation

$$s\alpha + \alpha_2 = r, \quad \alpha, \alpha_2 \in \mathcal{S},$$

where

$$\mathcal{S} = \left\{ \frac{t - t^{-1}}{2} : t \in \mathbb{Q}^\times \right\}.$$

The open diagonal segment corresponds to the interval

$$I = \left(-\frac{a}{b}, \frac{D-a}{b} \right).$$

The exceptional identity, together with $s < 0$, implies that $-s$ is a positive rational square. Write

$$s = -k^2, \quad k \in \mathbb{Q}_{>0}, \quad k \neq 1.$$

The identity gives

$$r = \varepsilon \frac{1 - k^4}{2k}, \quad \varepsilon \in \{\pm 1\}.$$

Replacing (α, α_2, r) by $(-\alpha, -\alpha_2, -r)$ reflects the interval I to $-I$ and preserves the number of rational detour solutions in the interval. Thus it is enough to prove the result in the sign convention

$$r = \frac{1 - k^4}{2k}.$$

The other sign follows by reflecting back. The special value is

$$\tau = \frac{r}{s-1} = \frac{k^2 - 1}{2k}.$$

Geometrically, $x = a + b\tau$ is the intersection point of the line AB with the x -axis. Because the quadrilateral is convex, the two diagonals meet in their interiors. Hence

$$\tau \in I.$$

Now consider the Case III genus-one curve attached to this exceptional identity. With

$$\alpha = \frac{t - t^{-1}}{2},$$

the condition $\alpha_2 = r - s\alpha \in \mathcal{S}$ gives a quartic C_k . Using the special point corresponding to $\alpha = \tau$ as the origin, C_k is birational to the elliptic curve

$$\begin{aligned} E_k : \quad Y^2 - 2(k^4 - 2k^2 - 1)XY + 8k^6(k^4 - 1)Y \\ = X^3 - 4k^4(k^2 + 1)X^2. \end{aligned}$$

The Case III torsion computation in Proposition 7.4 and Appendix A gives

$$E_k(\mathbb{Q})_{\text{tors}} \simeq \mathbb{Z}/8\mathbb{Z},$$

and, more importantly for the present argument, its torsion table shows that the only finite α -value represented by rational torsion is

$$\alpha = \tau.$$

The proof of that torsion table uses Mazur's torsion theorem and the Fermat–Mordell quartic case to rule out extra rational 2-torsion.

The two endpoint values of the interval I are

$$\lambda = -\frac{a}{b}, \quad \mu = \frac{D-a}{b}.$$

They are rational detour values because the endpoint distances

$$UA, UB, VA, VB$$

are the side lengths of the quadrilateral. Moreover neither endpoint U nor V lies on the line AB , since the quadrilateral is non-degenerate. Therefore neither endpoint value equals the intersection value:

$$\lambda \neq \tau, \quad \mu \neq \tau.$$

Consequently every rational point on the Case III curve with finite α -value different from τ is non-torsion. Hence the rational points corresponding to λ and μ are non-torsion.

The real points of an elliptic curve form a compact one-dimensional real Lie group, and the cyclic subgroup generated by a non-torsion point is dense in the real component it meets. Applying this density statement to either non-torsion endpoint point gives infinitely many rational points on the real component containing that endpoint. At the endpoint points used here the positive real detour branch is nonsingular, and the coordinate α is a real analytic local parameter. Hence every sufficiently small one-sided neighborhood of the relevant endpoint inside I contains infinitely many rational points of the Case III curve. Therefore there are infinitely many rational values

$$\alpha \in I$$

for which both α and $r - s\alpha$ lie in $\mathcal{S}$.

For each such α , set

$$x = a + b\alpha, \quad P = (x, 0).$$

Then $P \in UV^\circ$. Since $x \in \mathbb{Q}$ and $D \in \mathbb{Q}$, the distances

$$PU = x, \quad PV = D - x$$

are rational. The detour conditions

$$\alpha, r - s\alpha \in \mathcal{S}$$

give $PA, PB \in \mathbb{Q}$. Thus there are infinitely many rational points on UV° at rational distance from all four vertices. $\square$

Proposition 5.7 (The case $s = -1$, non-symmetric). *Assume the normalized convex quadrilateral setup. Suppose*

$$s = -1, \quad a + c \neq D. \tag{5.5}$$

Assume also that the side lengths

$$UA, AV, VB, BU$$

are rational. Then the open segment UV contains a rational point whose distances from U, V, A, B are rational.

Proof. The condition $s = -1$ means

$$d = -b.$$

The diagonal AB intersects the x -axis at

$$M = \left(\frac{a+c}{2}, 0 \right).$$

Convexity implies that M lies on the open segment UV . Hence

$$0 < \frac{a+c}{2} < D,$$

or equivalently

$$0 < a+c < 2D. \tag{5.6}$$

First suppose

$$a+c < D.$$

Set

$$P = (a+c, 0).$$

Then (5.6) gives $0 < a+c < D$, so P lies on the open segment UV . Moreover,

$$PA^2 = (a+c-a)^2 + b^2 = c^2 + b^2 = BU^2,$$

and

$$PB^2 = (a+c-c)^2 + b^2 = a^2 + b^2 = AU^2.$$

The distances BU and AU are side lengths of the normalized quadrilateral, hence rational. Therefore PA and PB are rational. Also

$$PU = a+c, \quad PV = D-a-c$$

are rational.

Now suppose

$$a+c > D.$$

Set

$$P = (a+c-D, 0).$$

Using (5.6), we have

$$0 < a+c-D < D,$$

so P lies on the open segment UV . This time,

$$PA^2 = (a+c-D-a)^2 + b^2 = (c-D)^2 + b^2 = BV^2,$$

and

$$PB^2 = (a+c-D-c)^2 + b^2 = (a-D)^2 + b^2 = AV^2.$$

The distances BV and AV are side lengths of the normalized quadrilateral, hence rational. Therefore PA and PB are rational, and

$$PU = a+c-D, \quad PV = 2D-a-c$$

are rational. The assumption $a+c \neq D$ leaves no other case. □

5.3 Infinitude in the convex non-symmetric $s = -1$ case

The next convex exceptional branch is the equal-height case $s = -1$. The symmetric subcase $a + c = D$ is treated later by the symmetric-parallelogram analysis and can have finite rank-zero examples. The non-symmetric subcase does not contribute finite examples.

Proposition 5.8 (Convex non-symmetric equal-height case gives infinitely many). *Assume the normalized convex quadrilateral setup*

$$U = (0, 0), \quad V = (D, 0), \quad A = (a, b), \quad B = (c, -b), \quad b > 0.$$

Assume

$$a + c \neq D.$$

Assume also that the side lengths

$$UA, \quad AV, \quad VB, \quad BU$$

are rational. Then the open diagonal segment UV° contains infinitely many rational points whose distances from U, V, A, B are rational.

Proof. For $P = (x, 0)$, put

$$\alpha = \frac{x - a}{b}.$$

Then $P \in UV^\circ$ if and only if

$$\alpha \in I := \left(-\frac{a}{b}, \frac{D - a}{b} \right).$$

Since $s = -1$, the detour equation has the form

$$\alpha_2 = \alpha - T,$$

where

$$T = -r = \frac{c - a}{b}.$$

If $T = 0$, then the condition becomes simply

$$\alpha \in \mathcal{S}.$$

Since $\mathcal{S}$ is dense in $\mathbb{R}$, the interval I contains infinitely many such rational values. For each of them,

$$P = (a + b\alpha, 0)$$

lies in UV° , has rational distances to A and B , and has rational distances to U and V . Hence the proposition holds in this case.

Assume from now on that $T \neq 0$. Thus the rational-distance condition from A and B is

$$\alpha \in \mathcal{S}, \quad \alpha - T \in \mathcal{S}.$$

The central value is

$$\tau = \frac{T}{2}.$$

Geometrically, $x = a + b\tau$ is the intersection of the line AB with the x -axis. Since the quadrilateral is convex, the two diagonals meet in their interiors, so

$$\tau \in I.$$

The genus-one curve for this simultaneous condition is birational to

$$E_T : Y^2 = X^3 + (T^2 + 2)X^2 + X.$$

The torsion-value calculation in Proposition 7.3 gives the following description of finite α -values represented by rational torsion points on E_T . The central value $T/2$ can occur, and if additional torsion values occur, they occur as a symmetric pair

$$\eta, \quad T - \eta,$$

one on each side of $T/2$. In particular, there is at most one torsion value on each side of $T/2$.

The endpoint values of I are

$$\lambda = -\frac{a}{b}, \quad \mu = \frac{D - a}{b}.$$

They are rational detour values because the four endpoint distances

$$UA, UB, VA, VB$$

are the side lengths of the quadrilateral. Since the quadrilateral is non-degenerate, neither endpoint lies on the line AB , and hence

$$\lambda \neq \frac{T}{2}, \quad \mu \neq \frac{T}{2}.$$

Suppose, for contradiction, that both endpoint points were torsion. Since $\tau = T/2$ lies strictly between λ and μ , the torsion-value description forces the two endpoint values to be the symmetric torsion pair:

$$\{\lambda, \mu\} = \{\eta, T - \eta\}.$$

Therefore

$$\lambda + \mu = T.$$

But

$$\lambda + \mu = -\frac{a}{b} + \frac{D - a}{b} = \frac{D - 2a}{b},$$

whereas

$$T = \frac{c - a}{b}.$$

Thus the equality $\lambda + \mu = T$ implies

$$D - 2a = c - a,$$

or equivalently

$$a + c = D.$$

This contradicts the non-symmetric hypothesis $a + c \neq D$.

Hence the two endpoint points cannot both be torsion. At least one endpoint point is non-torsion. By density of the cyclic subgroup generated by a non-torsion point on the relevant real elliptic component, the component containing that endpoint contains infinitely many rational multiples of the endpoint point. At the endpoint points used here the positive real detour branch is nonsingular, and the coordinate α is a real analytic local parameter. Therefore every sufficiently small one-sided neighborhood of the relevant endpoint inside I contains infinitely many rational detour values satisfying the simultaneous condition. For each such α , the point

$$P = (a + b\alpha, 0)$$

lies in UV° , has rational distances to A and B , and has rational distances to U and V because $x = a + b\alpha \in \mathbb{Q}$ and $D \in \mathbb{Q}$. Therefore there are infinitely many required points on UV° . $\square$

6 The symmetric convex parallelogram case

It remains to handle the exceptional situation

$$s = -1, \quad a + c = D. \quad (6.1)$$

Then $d = -b$ and $c = D - a$, so the normalized vertices are

$$U = (0, 0), \quad V = (D, 0), \quad A = (a, b), \quad B = (D - a, -b), \quad b > 0.$$

This is a parallelogram symmetric about the midpoint of UV .

Proposition 6.1 (Symmetric convex parallelogram case). *Assume the symmetric convex parallelogram setup above, and assume that the side lengths*

$$UA, \quad AV, \quad VB, \quad BU$$

are rational. Then the open segment UV contains a rational point whose distances from U, V, A, B are rational. If the other diagonal AB is not rational, then there are infinitely many such rational points.

Lemma 6.2 (Symmetric line reduction). *Assume the symmetric convex parallelogram setup. Put*

$$p = \frac{a}{b}, \quad q = \frac{D - a}{b}, \quad T = q - p = \frac{D - 2a}{b}.$$

For a point $P = (x, 0)$, set

$$\alpha = \frac{x - a}{b}.$$

Then

$$0 < x < D \iff -p < \alpha < q. \quad (6.2)$$

Moreover PA and PB are rational if and only if

$$1 + \alpha^2, \quad 1 + (\alpha - T)^2$$

are squares in $\mathbb{Q}$. If the other diagonal AB has rational length, then the midpoint of UV is a required point. If $AB \notin \mathbb{Q}$, then

$$T^2 + 4 \notin (\mathbb{Q}^\times)^2. \quad (6.3)$$

In particular $T \neq 0$.

Proof. For $P = (x, 0)$ and $\alpha = (x - a)/b$, we have

$$x = a + b\alpha.$$

Since $b > 0$, this gives (6.2). Also

$$PA^2 = (x - a)^2 + b^2 = b^2(1 + \alpha^2),$$

and

$$\begin{aligned} PB^2 &= (x - (D - a))^2 + b^2 \\ &= b^2(1 + (\alpha - T)^2), \end{aligned}$$

because

$$\alpha - T = \frac{x - a}{b} - \frac{D - 2a}{b} = \frac{x - (D - a)}{b}.$$

As $b \in \mathbb{Q}^\times$, the rationality of PA and PB is exactly the rational-square condition in the statement.

If $AB \in \mathbb{Q}$, the midpoint

$$P = \left(\frac{D}{2}, 0 \right)$$

is the common midpoint of the two diagonals of the parallelogram. Hence

$$PU = PV = \frac{D}{2}, \quad PA = PB = \frac{AB}{2},$$

so all four distances are rational.

Finally,

$$AB^2 = (D - 2a)^2 + (2b)^2 = b^2(T^2 + 4).$$

Since $b \in \mathbb{Q}^\times$, the assumption $AB \notin \mathbb{Q}$ is equivalent to (6.3). If $T = 0$, then $T^2 + 4 = 4$ is a rational square, contradicting (6.3). Thus $T \neq 0$. $\square$

Lemma 6.3 (The symmetric detour curve). *Assume $T \in \mathbb{Q}^\times$. The simultaneous rational-square conditions*

$$1 + \alpha^2 \in (\mathbb{Q}^\times)^2, \quad 1 + (\alpha - T)^2 \in (\mathbb{Q}^\times)^2$$

are represented by rational points on

$$\Gamma_T : (t - u)(tu + 1) = 2Ttu. \tag{6.4}$$

through

$$\alpha = \frac{t - t^{-1}}{2}, \quad \alpha - T = \frac{u - u^{-1}}{2}, \quad t, u \in \mathbb{Q}^\times.$$

The curve Γ_T is birational over $\mathbb{Q}$ to

$$E_T : Y^2 = X^3 + (T^2 + 2)X^2 + X. \tag{6.5}$$

The birational map $\phi : \Gamma_T \dashrightarrow E_T$ is

$$X = -1 - \frac{2T}{t - u - 2T}, \quad Y = \frac{T(t + u)}{t - u - 2T}, \tag{6.6}$$

and the inverse map, on $X \neq -1$, is

$$t = \frac{TX - Y}{X + 1}, \quad u = -\frac{TX + Y}{X + 1}. \tag{6.7}$$

Proof. By Lemma 2.4, the condition that $1 + \alpha^2$ is a rational square is equivalent to writing

$$\alpha = \frac{t - t^{-1}}{2}$$

for some $t \in \mathbb{Q}^\times$. Applying the same parametrization to $\alpha - T$ gives

$$\alpha - T = \frac{u - u^{-1}}{2}$$

with $u \in \mathbb{Q}^\times$. Subtracting these two equations gives (6.4). Conversely, any rational $(t, u) \in \Gamma_T$ with $t, u \neq 0$ gives such an α , and then

$$1 + \alpha^2 = \left(\frac{t + t^{-1}}{2} \right)^2, \quad 1 + (\alpha - T)^2 = \left(\frac{u + u^{-1}}{2} \right)^2.$$

The discriminant of E_T is

$$16T^2(T^2 + 4),$$

which is nonzero because $T \neq 0$ and $T^2 + 4 > 0$. Hence E_T is a nonsingular elliptic curve over $\mathbb{Q}$. A direct substitution of (6.6) gives

$$\begin{aligned} Y^2 - (X^3 + (T^2 + 2)X^2 + X) \\ = \frac{4T^2((t - u)(tu + 1) - 2Ttu)}{(t - u - 2T)^3}. \end{aligned} \quad (6.8)$$

Thus (6.6) sends Γ_T to E_T wherever it is defined. If $t - u - 2T = 0$, then $t - u = 2T$, and substituting in (6.4) gives

$$2T(tu + 1) = 2Ttu,$$

impossible because $T \neq 0$. Hence (6.6) is defined at every finite point of Γ_T .

Substituting (6.7) into (6.4), and using (6.5), gives the equation of Γ_T . The formulas (6.6) and (6.7) are inverse on the indicated open sets. Therefore Γ_T and E_T are birational over $\mathbb{Q}$. $\square$

Lemma 6.4 (Torsion in the nonrational-diagonal subcase). *Assume $T \in \mathbb{Q}^\times$ and*

$$T^2 + 4 \notin (\mathbb{Q}^\times)^2.$$

Then

$$E_T(\mathbb{Q})_{\text{tors}} \{ \mathcal{O}, (0, 0), (-1, T), (-1, -T) \}. \quad (6.9)$$

Proof. Write

$$A_T = T^2 + 2, \quad E_T : Y^2 = X^3 + A_T X^2 + X.$$

The point $(0, 0)$ has order 2. Also

$$(-1, \pm T) \in E_T(\mathbb{Q}).$$

At $(-1, T)$, the tangent slope is

$$m = \frac{3(-1)^2 + 2A_T(-1) + 1}{2T} \frac{4 - 2(T^2 + 2)}{2T} - T.$$

For a curve $Y^2 = X^3 + A_T X^2 + X$, duplication gives

$$X(2R) = m^2 - A_T - 2X(R).$$

Therefore

$$X(2(-1, T)) = T^2 - (T^2 + 2) + 2 = 0,$$

and the Y -coordinate is 0. Hence

$$2(-1, T) = (0, 0),$$

so $(-1, T)$ and $(-1, -T)$ have order 4.

The rational 2-torsion points are the rational roots of

$$X(X^2 + A_T X + 1).$$

The quadratic factor has discriminant

$$A_T^2 - 4 = (T^2 + 2)^2 - 4 = T^2(T^2 + 4),$$

which is not a rational square by hypothesis. Hence $(0, 0)$ is the only nontrivial rational 2-torsion point.

Mazur's theorem classifies the possible torsion subgroups of elliptic curves over $\mathbb{Q}$: the torsion subgroup is either cyclic of order $1, \dots, 10$ or 12 , or is isomorphic to $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2m\mathbb{Z}$ for $1 \leq m \leq 4$ [4]. Since $E_T(\mathbb{Q})$ contains a point of order 4 and only one nontrivial rational point of order 2, the only possibilities are

$$\mathbb{Z}/4\mathbb{Z}, \quad \mathbb{Z}/8\mathbb{Z}, \quad \mathbb{Z}/12\mathbb{Z}.$$

We exclude $\mathbb{Z}/8\mathbb{Z}$. Suppose $R = (X, Y) \in E_T(\mathbb{Q})$ doubled to one of the points $(-1, \pm T)$. Then $Y \neq 0$ and $X(2R) = -1$. The duplication formula gives

$$X(2R) + 1 = \frac{(X + 1)^4 + 4T^2 X^2}{4Y^2}. \quad (6.10)$$

If $X(2R) = -1$, then

$$(X + 1)^4 + 4T^2 X^2 = 0.$$

If $X \neq 0$, this gives

$$\left(\frac{(X + 1)^2}{2TX} \right)^2 = -1,$$

impossible over $\mathbb{Q}$. If $X = 0$, then the same equation gives $(X + 1)^4 = 1 \neq 0$. Thus $E_T(\mathbb{Q})$ has no point of order 8.

It remains to exclude rational 3-torsion. The 3-division polynomial of E_T is

$$\psi_3(X) = 3X^4 + 4(T^2 + 2)X^3 + 6X^2 - 1. \quad (6.11)$$

If E_T had a rational point of order 3, then its X -coordinate would be a rational root of ψ_3 . Rearranging $\psi_3(X) = 0$ gives

$$4T^2 X^3 = (1 - 3X)(X + 1)^3. \quad (6.12)$$

The values $X = 0$ and $X = -1$ do not satisfy (6.11), so they are excluded. Set

$$Z = 3X, \quad W = \frac{6TX^2}{X+1}.$$

Multiplying (6.12) by $9X/(X+1)^2$, we obtain

$$W^2 = Z(1-Z)(Z+3). \quad (6.13)$$

With the change of variables

$$x = -Z - 1, \quad y = W,$$

equation (6.13) becomes

$$C: \quad y^2 = x^3 + x^2 - 4x - 4 = (x+2)(x+1)(x-2). \quad (6.14)$$

The LMFDB entry 48.a4, also Cremona label 48a1, together with Cremona's tables, gives the following result for this curve [2, 7]:

$$\text{rank } C(\mathbb{Q}) = 0,$$

and

$$C(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

This cited database is used only for this fixed curve C . Since $\text{rank } C(\mathbb{Q}) = 0$, all rational points of C are torsion. Since the cubic in (6.14) splits completely, the four rational torsion points are exactly

$$\mathcal{O}, \quad (-2, 0), \quad (-1, 0), \quad (2, 0).$$

Undoing $x = -Z - 1$, the affine points in this list give

$$Z = 1, \quad Z = 0, \quad Z = -3.$$

Thus the possible original X -coordinates are

$$X = \frac{1}{3}, \quad X = 0, \quad X = -1.$$

The values 0 and -1 have already been excluded. If $X = 1/3$, then the right side of (6.12) is zero, so $T = 0$, contradicting the hypothesis. Therefore $E_T(\mathbb{Q})$ has no point of order 3, and hence the torsion group in the list above cannot be $\mathbb{Z}/12\mathbb{Z}$. Consequently (6.9) holds. $\square$

Lemma 6.5 (The endpoint point is non-torsion). *Assume the symmetric convex parallelogram setup, assume the four side lengths are rational, and assume $AB \notin \mathbb{Q}$. Let*

$$h_p = \sqrt{1+p^2}, \quad h_q = \sqrt{1+q^2},$$

and put

$$Q_0 = (t_0, u_0) := (h_p - p, h_q - q) \in \Gamma_T(\mathbb{Q}).$$

Let $R_0 = \phi(Q_0) \in E_T(\mathbb{Q})$, where ϕ is the birational map in (6.6). Then R_0 is non-torsion.

Proof. In the symmetric convex parallelogram, $UB = AV$ and $VB = UA$. The side-length hypothesis therefore implies that UA and UB are rational. Since

$$UA^2 = a^2 + b^2 = b^2(1 + p^2),$$

and

$$UB^2 = (D - a)^2 + b^2 = b^2(1 + q^2),$$

we have

$$h_p = \frac{UA}{b} \in \mathbb{Q}_{>0}, \quad h_q = \frac{UB}{b} \in \mathbb{Q}_{>0}.$$

Thus $t_0, u_0 \in \mathbb{Q}$. Moreover $t_0, u_0 > 0$, because $\sqrt{1 + z^2} > z$ for every real number z . The standard parametrization then gives

$$\frac{t_0 - t_0^{-1}}{2} = -p, \quad \frac{u_0 - u_0^{-1}}{2} = -q.$$

Since $-p - T = -q$, the point Q_0 lies on $\Gamma_T(\mathbb{Q})$.

By Lemma 6.2, the assumption $AB \notin \mathbb{Q}$ implies

$$T^2 + 4 \notin (\mathbb{Q}^\times)^2,$$

so Lemma 6.4 applies. First R_0 is finite. If

$$t_0 - u_0 - 2T = 0,$$

then $t_0 - u_0 = 2T$, and substituting into Γ_T gives

$$2T(t_0 u_0 + 1) = 2T t_0 u_0,$$

impossible because $T \neq 0$. Thus $R_0 \notin \mathcal{O}$.

By (6.9), to prove that R_0 is non-torsion it remains to exclude the three finite torsion points. Its X -coordinate is

$$X(R_0) = -1 - \frac{2T}{t_0 - u_0 - 2T},$$

which is not -1 because $T \neq 0$. Hence

$$R_0 \neq (-1, T), \quad R_0 \neq (-1, -T).$$

Finally,

$$Y(R_0) = \frac{T(t_0 + u_0)}{t_0 - u_0 - 2T}.$$

Here $T \neq 0$, $t_0 > 0$, $u_0 > 0$, and the denominator is nonzero, so

$$Y(R_0) \neq 0.$$

Thus $R_0 \neq (0, 0)$. Therefore $R_0 \in E_T(\mathbb{Q})$ is non-torsion. □

Proof of Proposition 6.1. Use the notation of Lemma 6.2. If $AB \in \mathbb{Q}$, that lemma shows that the midpoint of UV is a required point. Assume therefore that $AB \notin \mathbb{Q}$. Lemma 6.2 gives $T \neq 0$ and (6.3), Lemma 6.3 gives the elliptic model, and Lemma 6.5 gives a non-torsion point $R_0 \in E_T(\mathbb{Q})$ lying over the endpoint U .

We now use density of multiples on the real elliptic curve. The real points $E_T(\mathbb{R})$ form a compact one-dimensional real Lie group. Its identity component is isomorphic to $\mathbb{R}/\mathbb{Z}$, and a non-torsion element corresponds to an irrational rotation. Therefore the cyclic subgroup generated by a non-torsion point is dense in the real component it meets. If the point is not in the identity component, then its odd multiples are dense in its component, because twice the point lies in the identity component.

For real α near $-p$, define

$$t(\alpha) = \sqrt{1 + \alpha^2} + \alpha, \quad u(\alpha) = \sqrt{1 + (\alpha - T)^2} + \alpha - T,$$

using the positive square roots. Then $t(\alpha)$ and $u(\alpha)$ are positive real analytic functions, and

$$\alpha = \frac{t(\alpha) - t(\alpha)^{-1}}{2}, \quad \alpha - T = \frac{u(\alpha) - u(\alpha)^{-1}}{2}.$$

Thus

$$Q(\alpha) = (t(\alpha), u(\alpha)) \in \Gamma_T(\mathbb{R}),$$

and $Q(-p) = Q_0$.

Choose $\varepsilon > 0$ small enough that

$$J = (-p, -p + \varepsilon) \subset (-p, q),$$

and that the inverse formula (6.7) is valid on $\phi(Q(J))$. Then

$$\phi(Q(J))$$

is a nonempty open real arc in the connected component of $E_T(\mathbb{R})$ containing R_0 . Since R_0 is non-torsion, infinitely many multiples of R_0 , in the appropriate parity class if necessary, lie on this arc.

For one such multiple $R_n = nR_0$, use the inverse map (6.7). We obtain rational numbers $t_n, u_n \in \mathbb{Q}^\times$ and some $\alpha_n \in J$ such that

$$\alpha_n = \frac{t_n - t_n^{-1}}{2}, \quad \alpha_n - T = \frac{u_n - u_n^{-1}}{2}.$$

In particular $\alpha_n \in \mathbb{Q}$. Set

$$x_n = a + b\alpha_n, \quad P_n = (x_n, 0).$$

Because $\alpha_n \in J \subset (-p, q)$, equation (6.2) gives

$$0 < x_n < D.$$

Thus P_n lies on the open segment UV . The distances to U and V are

$$P_n U = x_n, \quad P_n V = D - x_n,$$

which are rational. Also

$$1 + \alpha_n^2 = \left(\frac{t_n + t_n^{-1}}{2} \right)^2,$$

and

$$1 + (\alpha_n - T)^2 = \left(\frac{u_n + u_n^{-1}}{2} \right)^2.$$

Therefore

$$P_n A = b \left| \frac{t_n + t_n^{-1}}{2} \right| \in \mathbb{Q},$$

and

$$P_n B = b \left| \frac{u_n + u_n^{-1}}{2} \right| \in \mathbb{Q}.$$

This gives a required point. Since infinitely many distinct multiples of the non-torsion point R_0 meet the arc, and the birational inverse is injective on the chosen arc, the construction gives infinitely many distinct points P_n on UV . This proves the proposition. $\square$

6.1 Rank-zero criterion for symmetric convex parallelograms

Consider the normalized symmetric convex parallelogram

$$U = (0, 0), \quad V = (D, 0),$$

$$A = (a, b), \quad B = (D - a, -b), \quad a, b, D \in \mathbb{Q}, \quad b > 0.$$

Put

$$p = \frac{a}{b}, \quad q = \frac{D - a}{b}, \quad T = q - p = \frac{D - 2a}{b}.$$

For $P = (x, 0)$, put

$$\alpha = \frac{x - a}{b}.$$

Then

$$x = a + b\alpha,$$

and therefore

$$P \in UV^\circ \iff -p < \alpha < q.$$

Furthermore

$$PA^2 = b^2(1 + \alpha^2),$$

and

$$PB^2 = b^2(1 + (\alpha - T)^2).$$

Thus P has rational distance from A and B if and only if

$$1 + \alpha^2, \quad 1 + (\alpha - T)^2$$

are rational squares.

Parametrize the two rational-square conditions by

$$\alpha = \frac{t - t^{-1}}{2}, \quad \alpha - T = \frac{u - u^{-1}}{2}, \quad t, u \in \mathbb{Q}^\times.$$

Eliminating α gives the affine curve

$$\Gamma_T : (t - u)(tu + 1) = 2Ttu.$$

For $T \neq 0$, this curve is birational over $\mathbb{Q}$ to

$$E_T : Y^2 = X^3 + (T^2 + 2)X^2 + X,$$

via

$$X = -1 - \frac{2T}{t - u - 2T}, \quad Y = \frac{T(t + u)}{t - u - 2T},$$

with inverse

$$t = \frac{TX - Y}{X + 1}, \quad u = -\frac{TX + Y}{X + 1}.$$

The denominator $t - u - 2T$ is nonzero on Γ_T : if $t - u - 2T = 0$, then $t - u = 2T$, and substituting in the defining equation of Γ_T gives

$$2T(tu + 1) = 2Ttu,$$

contradicting $T \neq 0$.

Proposition 6.6 (Rank-zero finiteness criterion). *Assume $T \neq 0$. If $E_T(\mathbb{Q})$ has rank 0, then $\mathcal{R}(U, V; A, B)$, and hence $\mathcal{R}^\circ(U, V; A, B)$, is finite.*

Proof. Every point $P = (x, 0) \in \mathcal{R}(U, V; A, B)$ gives a rational value $\alpha = (x - a)/b$. The conditions $PA, PB \in \mathbb{Q}$ give rational $t, u \in \mathbb{Q}^\times$ satisfying the defining equation of Γ_T , after choosing signs in the standard parametrization of $1 + \alpha^2$ and $1 + (\alpha - T)^2$. Hence every such P gives a rational point on Γ_T , and therefore, by the birational map, a rational point on E_T .

If $E_T(\mathbb{Q})$ has rank 0, then $E_T(\mathbb{Q})$ is its finite torsion group. Thus there are only finitely many rational points on E_T , hence only finitely many rational points on Γ_T , hence only finitely many possible values of α , and therefore only finitely many points $P = (a + b\alpha, 0)$ on UV . $\square$

6.2 A finite symmetric convex parallelogram example

Example 6.7. Let

$$U = (0, 0), \quad A = (18, 24), \quad V = (50, 0), \quad B = (32, -24).$$

Then $UAVB$ is a symmetric convex parallelogram. Its side lengths are

$$UA = VB = 30, \quad AV = BU = 40,$$

and its diagonals have lengths

$$UV = 50, \quad AB = 50.$$

Thus this is an integer-coordinate parallelogram with integer side lengths and integer diagonals.

Theorem 6.8 (A rank-zero finite symmetric convex parallelogram example). *For the parallelogram in Example 6.7,*

$$\mathcal{R}^\circ(U, V; A, B) = \{(25, 0)\}.$$

Equivalently, the only rational point strictly between U and V whose distances from all four vertices are rational is the midpoint

$$P = (25, 0).$$

Proof. Here

$$D = 50, \quad a = 18, \quad b = 24.$$

Hence

$$p = \frac{a}{b} = \frac{3}{4}, \quad q = \frac{D-a}{b} = \frac{4}{3},$$

and

$$T = q - p = \frac{4}{3} - \frac{3}{4} = \frac{7}{12}.$$

For $P = (x, 0)$, put

$$\alpha = \frac{x - 18}{24}.$$

Then P lies strictly between U and V if and only if

$$-\frac{3}{4} < \alpha < \frac{4}{3}.$$

The corresponding elliptic curve is

$$E_{7/12} : Y^2 = X^3 + \frac{337}{144}X^2 + X.$$

With

$$x = 144X, \quad y = 1728Y,$$

this becomes

$$E : y^2 = x^3 + 337x^2 + 20736x = x(x + 81)(x + 256).$$

We use the standard Cremona–LMFDB rank and torsion results for this curve [2, 6]: it has rank 0, and its rational torsion subgroup is isomorphic to

$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}.$$

Therefore every rational point on $E_{7/12}$ is torsion.

For this particular value $T = 7/12$, the rational torsion points on $E_{7/12}$ are

$$\begin{aligned} \mathcal{O}, \quad (0, 0), \quad \left(-1, \pm \frac{7}{12}\right), \quad \left(1, \pm \frac{25}{12}\right), \\ \left(-\frac{9}{16}, 0\right), \quad \left(-\frac{16}{9}, 0\right), \\ \left(6, \pm \frac{35}{2}\right), \quad \left(\frac{1}{6}, \pm \frac{35}{72}\right), \quad \left(-\frac{2}{3}, \pm \frac{5}{18}\right), \quad \left(-\frac{3}{2}, \pm \frac{5}{8}\right). \end{aligned}$$

There are 16 points in this list, which equals the order of $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$, so the list is exhaustive. Substituting these points in the inverse formula

$$t = \frac{TX - Y}{X + 1}, \quad \alpha = \frac{t - t^{-1}}{2},$$

where the formula is defined and $t \neq 0$, gives exactly the following finite α -values:

α	points of $E_{7/12}(\mathbb{Q})$ producing this value
$-\frac{3}{4}$	$(6, \frac{35}{2}), (\frac{1}{6}, -\frac{35}{72}), (-\frac{2}{3}, \frac{5}{18}), (-\frac{3}{2}, -\frac{5}{8})$
$\frac{7}{24}$	$(1, \pm\frac{25}{12}), (-\frac{9}{16}, 0), (-\frac{16}{9}, 0)$
$\frac{4}{3}$	$(6, -\frac{35}{2}), (\frac{1}{6}, \frac{35}{72}), (-\frac{2}{3}, -\frac{5}{18}), (-\frac{3}{2}, \frac{5}{8})$

The remaining torsion points give no finite admissible α -value: $\mathcal{O}$ is the point at infinity, $(0, 0)$ gives $t = 0$, and $(-1, \pm 7/12)$ lie outside the affine inverse chart $X \neq -1$. These chart exclusions do not hide additional finite values. Every finite point of Γ_T maps by the formula to an affine point of E_T , so it cannot map to $\mathcal{O}$. If a finite point mapped to $(0, 0)$, the inverse formula, which is valid at $X = 0$, would give $t = u = 0$, contradicting $t, u \in \mathbb{Q}^\times$. If a finite point mapped to one of the points with $X = -1$, then

$$X = -1 - \frac{2T}{t - u - 2T}$$

would force $T = 0$, whereas here $T = 7/12$.

Thus the only possible rational values of α are

$$-\frac{3}{4}, \quad \frac{7}{24}, \quad \frac{4}{3}.$$

The values $-3/4$ and $4/3$ are precisely the endpoints of the interval. Indeed,

$$\alpha = -\frac{3}{4} \iff x = 18 + 24 \left(-\frac{3}{4}\right) = 0,$$

and

$$\alpha = \frac{4}{3} \iff x = 18 + 24 \left(\frac{4}{3}\right) = 50.$$

The only value strictly inside the interval is

$$\alpha = \frac{7}{24}.$$

It gives

$$x = 18 + 24 \cdot \frac{7}{24} = 25.$$

Therefore

$$\mathcal{R}^\circ(U, V; A, B) = \{(25, 0)\}.$$

Finally, the point $P = (25, 0)$ indeed works:

$$PU = PV = 25,$$

and since P is also the midpoint of the other diagonal,

$$PA = PB = \frac{AB}{2} = 25.$$

This proves both existence and uniqueness of the interior point on the chosen diagonal. $\square$

Remark 6.9 (Symmetric convex parallelogram with rational AB can be finite). The conclusion of Theorem 6.8 is stronger than the existence theorem used in Proposition 6.1. That proposition only needs the midpoint. The uniqueness statement uses the rank-zero case above, and should therefore be understood as an arithmetic classification for this particular elliptic curve.

Thus Theorem 6.8 gives an explicit affirmative answer to the question whether the convex symmetric parallelogram case with rational other diagonal AB can have only finitely many rational points on the chosen diagonal. In this example $AB = 50$, and the only rational point in the open diagonal segment at rational distance from all four vertices is the midpoint $(25, 0)$. Therefore the rationality of the other diagonal AB should not be interpreted as an infinite-points condition; it gives an immediate midpoint solution, while infinitude depends on the rank of the associated curve.

7 Exceptional endpoint-line cases

Throughout this section we return to the normalized endpoint-line setup of Section 3. These endpoint arguments are used in the final proof for concave quadrilaterals, where the intersection point of the line AB with the diagonal line need not lie on the open diagonal segment.

Since $s < 0$, the exceptional alternatives not covered by Proposition 4.2 are

$$r = 0, \quad s = -1, \quad 4r^2s = -(1 - s^2)^2. \quad (7.1)$$

The proof of these three alternatives is unified. In each case there is a special rational value τ of α_1 . In all three cases this special value is

$$\tau = \frac{r}{s - 1}.$$

It has a simple geometric meaning: $x = a + b\tau$ is the intersection of the line AB with the x -axis. Indeed

$$r - s\tau = -\tau$$

is equivalent to $\tau = r/(s - 1)$. Since

$$\alpha_1 = \frac{x - a}{b}, \quad \alpha_2 = \frac{c - x}{d},$$

the equality $\alpha_2 = -\alpha_1$ says exactly that the points (a, b) , (c, d) , and $(x, 0)$ are collinear. Hence the hypothesis that neither endpoint U, V lies on AB implies

$$\lambda \neq \tau, \quad \mu \neq \tau.$$

In the first and third exceptional cases the special value itself satisfies $\tau, -\tau \in \mathcal{S}$, so if $\tau \in I$ it gives the desired point directly. In the second exceptional case $s = -1$, the special value is the central value $T/2$; it need not lie in $\mathcal{S}$ for arbitrary T , and the proof below uses the torsion-value

calculation instead. If the relevant direct point is not available, then the two endpoint values λ, μ lie strictly on the same side of the special value, or else the torsion-value calculation supplies the middle value $T/2$ as an available detour value. In the remaining endpoint cases, the torsion-value computation for the corresponding elliptic curve says that on either side of the special value there is at most one torsion value of α_1 . Since $\lambda \neq \mu$, at least one endpoint point is non-torsion. Density of its multiples on the real component then supplies infinitely many rational solutions with $\alpha_1 \in I$.

Lemma 7.1 (Density from one non-torsion endpoint). *Let Γ be one of the genus-one curves occurring below, equipped with a rational coordinate α . Suppose that the positive real branch of $\Gamma(\mathbb{R})$ is parameterized continuously by $\alpha \in \mathbb{R}$. Let $\lambda < \mu$ be two rational values on this branch, and suppose that at least one of the two points of $\Gamma(\mathbb{Q})$ over λ, μ is non-torsion for some elliptic curve group law on Γ . Then there are infinitely many rational points of Γ whose α -coordinates lie in (λ, μ) .*

Proof. Let R be a non-torsion endpoint point. If R lies over λ , use a right-hand neighborhood of λ ; if R lies over μ , use a left-hand neighborhood of μ . Let E be an elliptic curve birational to Γ , with the group law used to define torsion, and identify R with its image on E . The real Lie group $E(\mathbb{R})$ has either one or two connected components. On the identity component, a non-torsion point generates a dense cyclic subgroup. If R lies in the other component, then $2R$ is a non-torsion point in the identity component, so the even multiples of R are dense in the identity component and the odd multiples of R are dense in the component containing R . Thus, in all cases, infinitely many rational multiples of R lie in every nonempty open arc of the real component containing R .

The coordinate α is a real analytic local parameter on the positive branch at every finite point used below. Hence every sufficiently small one-sided neighborhood of the relevant endpoint on that branch contains infinitely many rational multiples of R . Taking the neighborhood small enough to lie inside (λ, μ) proves the claim. $\square$

7.1 First exceptional case: $r = 0$

Proposition 7.2 (The aligned case). *Assume the normalized endpoint-line setup and $r = 0$. Then the open segment UV contains a rational point P with $PA, PB \in \mathbb{Q}$. Writing $\rho = -s$, if $\rho = 1$, then there are infinitely many such points. If $\rho \neq 1$ and the endpoint points are not both torsion on the associated genus-one curve, then there are infinitely many such points.*

Proof. Since $r = 0$, the detour equation is

$$\alpha_2 = -s\alpha_1.$$

Put

$$\rho = -s > 0.$$

Thus the required condition is

$$\alpha_1 \in \mathcal{S}, \quad \rho\alpha_1 \in \mathcal{S}. \tag{7.2}$$

First suppose $\rho = 1$. Then (7.2) reduces simply to $\alpha_1 \in \mathcal{S}$. By Lemma 2.4, $\mathcal{S}$ is dense in $\mathbb{R}$, so I contains infinitely many rational values $\alpha_1 \in \mathcal{S}$. Lemma 3.2 gives the desired points.

Now suppose $\rho \neq 1$. The special value is

$$\tau = 0.$$

Parametrize the first condition $\alpha_1 \in \mathcal{S}$ by

$$\alpha_1 = \frac{t - t^{-1}}{2}.$$

The second condition $\rho\alpha_1 \in \mathcal{S}$ gives the Jacobi quartic

$$C_\rho : \quad W^2 = t^4 + \left(\frac{4}{\rho^2} - 2 \right) t^2 + 1. \quad (7.3)$$

This is a nonsingular genus-one curve because $\rho \neq 0, 1$. Its standard Jacobi-quartic elliptic model is

$$E_\rho : \quad Y^2 = X(X + \rho^2)(X + 1). \quad (7.4)$$

The coordinate α_1 is the common-leg coordinate on this model. We spell out the identification with the concordant-form notation. For $h, k \in \mathbb{Q}^\times$, the system

$$h^2 + z^2 = Z^2, \quad k^2 + z^2 = W^2$$

is the affine chart $X_1 = 1$, $X_0 = z$, of the intersection of quadrics which Selder and Spindler denote by $Q(h^2, k^2)$. Their Theorem 2 gives an explicit isomorphism between this intersection and

$$y^2 = x(x + h^2)(x + k^2),$$

and their inverse map sends a point of the elliptic curve back to the quadric coordinates. In the affine chart used here, the corresponding finite common-leg value is

$$z = \frac{X_0}{X_1}.$$

Reading the torsion points listed in Selder–Spindler’s Theorem 4 through this inverse map gives the following torsion values:

$$z = 0, \quad \text{and, if } hk \in (\mathbb{Q}^\times)^2, \quad z = \pm\sqrt{hk}.$$

The points of order 2 give the trivial value $z = 0$. In this square-coefficient subfamily the point $(0, 0)$ is divisible by 2, because the differences from the other two roots are h^2 and k^2 , both rational squares. Hence the curve has a rational point of order 4. Moreover the three roots $0, -h^2, -k^2$ are rational, so the curve has full rational 2-torsion. Mazur’s torsion theorem [4] then restricts the torsion subgroup to one of the groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2m\mathbb{Z}$, $1 \leq m \leq 4$, and the presence of a point of order 4 leaves only $m = 2$ or $m = 4$. Thus the 3- and 6-torsion cases in Selder–Spindler’s general list cannot occur in this subfamily. The possible 4- and 8-torsion points give exactly the values shown above. This is precisely the torsion-solution classification in [8, Theorem 2 and Theorem 4]; see also their discussion of concordant forms in Section 5.1.

Now apply this with

$$h = 1, \quad k = \frac{1}{\rho}, \quad z = \alpha_1.$$

Indeed the second equation $1 + \rho^2\alpha_1^2 = \square$ is equivalent, after division by ρ^2 , to

$$\frac{1}{\rho^2} + \alpha_1^2 = \square.$$

Thus the rational torsion points of E_ρ can give only

$$\alpha_1 = 0 \quad \text{and, when } \rho \text{ is a square in } \mathbb{Q}, \quad \alpha_1 = \pm \frac{1}{\sqrt{\rho}}.$$

If $0 \in I$, then $\alpha_1 = 0$ gives $\alpha_2 = 0$. Since $0 \in \mathcal{S}$, Lemma 3.2 gives the desired point. Moreover, if the two endpoint points are not both torsion on C_ρ , then at least one endpoint point is non-torsion. The endpoint values lie on the positive real branch, and the coordinate α_1 is a real analytic local parameter at those finite endpoint points. Lemma 7.1 therefore gives infinitely many rational solutions of (7.2) with $\alpha_1 \in I$. Thus the asserted infinitude also holds in the subcase $0 \in I$ whenever an endpoint point is non-torsion.

The torsion-value list above contains at most one positive value and at most one negative value. Assume now that $0 \notin I$. The endpoint values λ, μ are distinct and lie on the same side of 0. Since $\lambda \neq \mu$, at least one endpoint point on $C_\rho(\mathbb{Q})$ is non-torsion. Lemma 7.1 then gives infinitely many rational solutions of (7.2) with $\alpha_1 \in I$. Lemma 3.2 converts them to points P on UV with $PA, PB \in \mathbb{Q}$. $\square$

7.2 Second exceptional case: $s = -1$

Proposition 7.3 (The opposite equal-height case). *Assume the normalized endpoint-line setup, $s = -1$, and $r \neq 0$. Then the open segment UV contains a rational point P with $PA, PB \in \mathbb{Q}$. If the endpoint points are not both torsion on the associated genus-one curve, then there are infinitely many such points.*

Proof. Put

$$T = -r \in \mathbb{Q}^\times.$$

The detour equation becomes

$$\alpha_2 = \alpha_1 - T.$$

Thus we need

$$\alpha_1 \in \mathcal{S}, \quad \alpha_1 - T \in \mathcal{S}. \tag{7.5}$$

The special value is

$$\tau = \frac{T}{2} = -\frac{r}{2}.$$

Parametrize

$$\alpha_1 = \frac{t - t^{-1}}{2}, \quad \alpha_1 - T = \frac{u - u^{-1}}{2}.$$

Eliminating α_1 gives

$$(t - u)(tu + 1) = 2Ttu. \tag{7.6}$$

The curve (7.6) is birational to the nonsingular elliptic curve

$$E_T : Y^2 = X^3 + (T^2 + 2)X^2 + X. \tag{7.7}$$

whose discriminant is

$$16T^2(T^2 + 4) \neq 0.$$

The birational map is

$$X = -1 - \frac{2T}{t - u - 2T}, \quad Y = \frac{T(t + u)}{t - u - 2T},$$

with inverse

$$t = \frac{TX - Y}{X + 1}, \quad u = -\frac{TX + Y}{X + 1}. \quad (7.8)$$

Substitution verifies both directions wherever the denominators are nonzero. The denominator $t - u - 2T$ is in fact nonzero on the affine detour curve: if $t - u - 2T = 0$, then $t - u = 2T$, and (7.6) gives

$$2T(tu + 1) = 2Ttu,$$

contradicting $T \neq 0$.

We now determine the finite values of α_1 which can be represented by rational torsion points on E_T . Put $A = T^2 + 2$, so

$$E_T : Y^2 = X^3 + AX^2 + X.$$

The point

$$P_4 = (-1, T)$$

has exact order 4, and $2P_4 = (0, 0)$. We first rule out rational points of order 3. The third division polynomial of $Y^2 = X^3 + AX^2 + X$ is

$$\psi_3(X) = 3X^4 + 4AX^3 + 6X^2 - 1.$$

If $\psi_3(X) = 0$ for some $X \in \mathbb{Q}$, then

$$4T^2X^3 = -(X + 1)^3(3X - 1).$$

The value $X = -1$ is impossible, since

$$\psi_3(-1) = -4T^2 \neq 0.$$

Therefore $X + 1 \neq 0$. Since $T \neq 0$, putting

$$U = -3X, \quad V = \frac{6TX^2}{X + 1}$$

gives a rational point on

$$F : V^2 = U(U - 3)(U + 1).$$

The curve F is the Cremona curve 48.a4 after the integral change of variables

$$x = 9 \left(U - \frac{2}{3} \right), \quad y = 27V,$$

which gives

$$y^2 = x^3 - 351x - 1890.$$

Cremona's tables, equivalently the LMFDB entry 48.a4, give the following result [2, 7]:

$$F(\mathbb{Q}) = \{\mathcal{O}, (-1, 0), (0, 0), (3, 0)\} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

Thus $X \in \{1/3, 0, -1\}$. Substitution into ψ_3 gives, respectively,

$$\psi_3(1/3) = \frac{4T^2}{27}, \quad \psi_3(0) = -1, \quad \psi_3(-1) = -4T^2,$$

so none is zero. Hence $E_T(\mathbb{Q})$ has no rational 3-torsion.

By Mazur's torsion theorem [4], and since $E_T(\mathbb{Q})$ contains a point of order 4, the only remaining possibilities are

$$\mathbb{Z}/4\mathbb{Z}, \quad \mathbb{Z}/8\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}.$$

The cyclic order-8 case cannot occur. Indeed, if a rational point (X, Y) satisfied $2(X, Y) = (-1, \pm T)$, then the X -coordinate duplication formula would give

$$(X + 1)^4 + 4T^2X^2 = 0,$$

which is impossible over $\mathbb{Q}$. Indeed, if $X \neq 0$, then

$$\left(\frac{(X + 1)^2}{2TX} \right)^2 = -1,$$

while if $X = 0$, the left side is 1. Therefore

$$E_T(\mathbb{Q})_{\text{tors}} \in \{\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}\}.$$

Extra rational 2-torsion occurs exactly when

$$T^2 + 4 = w^2 \quad (w \in \mathbb{Q}).$$

If this condition fails, then the torsion points are only $\mathcal{O}$, $(0, 0)$, and $(-1, \pm T)$. The point $\mathcal{O}$ gives no finite detour value, because every finite detour point has $t - u - 2T \neq 0$ and hence maps by the formula to an affine point of E_T . The remaining torsion points also give no finite value of α_1 : the point $(0, 0)$ gives $t = 0$ in (7.8), while $(-1, \pm T)$ makes the denominator $X + 1$ vanish. These chart exclusions do not hide any finite detour value. If a finite detour point mapped to $(0, 0)$, then the inverse formula (7.8), which is valid at $X = 0$, would give $t = u = 0$, impossible because $t, u \in \mathbb{Q}^\times$. If a finite detour point mapped to $X = -1$, then the forward formula would force

$$\frac{-2T}{t - u - 2T} = 0,$$

again impossible because $T \neq 0$.

Assume now that $T^2 + 4 = w^2$. The two additional nonzero 2-torsion points have $Y = 0$ and satisfy

$$X^2 + (T^2 + 2)X + 1 = 0.$$

For such a point, (7.8) gives $t = TX/(X + 1)$. Using the quadratic equation for X , one obtains

$$t - \frac{1}{t} = T,$$

so these 2-torsion points give $\alpha_1 = T/2$. The points

$$(1, \pm w)$$

also have order 4, and (7.8) gives

$$t = \frac{T \mp w}{2}, \quad t - \frac{1}{t} = T;$$

hence they also give $\alpha_1 = T/2$.

It remains only to inspect possible points of order 8, which can occur only in the last case in the torsion list. Since the points $(-1, \pm T)$ are not divisible by 2 over $\mathbb{Q}$, every order-8 torsion point must be a half of one of $(1, \pm w)$. Solving the duplication equations for $2(X, Y) = (1, w)$ gives

$$X^2 - 2(1+w)X + 1 = 0 \quad \text{or} \quad X^2 - 2(1-w)X + 1 = 0.$$

For the first quadratic one has $Y = (w+1)X - 1$, and for the second one has $Y = (w-1)X + 1$. Substitution in (7.8) gives in both cases

$$\alpha_1 = \frac{T-w}{2}.$$

Replacing w by $-w$, which is the same calculation for halves of $(1, -w)$, gives

$$\alpha_1 = \frac{T+w}{2}.$$

Consequently the finite α_1 -values represented by rational torsion points are contained in

$$\frac{T}{2} \quad \text{and possibly a pair} \quad \eta, T - \eta.$$

Here, when the extra pair occurs, it is explicitly

$$\eta = \frac{T+w}{2}, \quad T - \eta = \frac{T-w}{2}.$$

The two values in this pair lie on opposite sides of $T/2$. Hence there is at most one torsion value on either side of $T/2$.

The value $T/2$ is the value of α_1 at the intersection of the line AB with the x -axis. By the endpoint nondegeneracy hypothesis, neither endpoint value is equal to $T/2$.

If at least one endpoint point is non-torsion, then Lemma 7.1 gives infinitely many rational solutions of (7.5) with $\alpha_1 \in I$, and Lemma 3.2 gives the required points P .

It remains to consider the case in which both endpoint points are torsion. If the endpoint values lie on the same side of $T/2$, this is impossible, because there is at most one torsion value on that side. Hence the endpoint values lie on opposite sides of $T/2$. By the torsion-value description above, the only way this can happen is that the endpoint values are the pair $\eta, T - \eta$ above. In that case $T/2$ is also a rational torsion value on the same detour curve, so

$$\frac{T}{2} \in \mathcal{S}, \quad -\frac{T}{2} \in \mathcal{S}.$$

Since $T/2$ lies strictly between η and $T - \eta$, it lies in I . Lemma 3.2 gives the required point P . $\square$

7.3 Third exceptional case: the quadratic exceptional identity

Proposition 7.4 (The quadratic exceptional identity). *Assume the normalized endpoint-line setup,*

$$r \neq 0, \quad s \neq -1, \quad 4r^2s = -(1 - s^2)^2.$$

Then the open segment UV contains a rational point P with $PA, PB \in \mathbb{Q}$. If the special value

$$\tau = \frac{r}{s-1}$$

does not lie in I , then there are infinitely many such points.

Proof. The exceptional identity and the assumptions $s < 0$, $r \neq 0$ imply

$$-s = \left(\frac{1 - s^2}{2r} \right)^2.$$

Thus $-s$ is a positive rational square. Write

$$s = -k^2, \quad k \in \mathbb{Q}_{>0}.$$

Since $s \neq -1$, we have $k \neq 1$. The exceptional identity gives

$$r = \varepsilon \frac{1 - k^4}{2k}, \quad \varepsilon \in \{\pm 1\}.$$

The sign of r is immaterial. Indeed, if

$$s\alpha_1 + \alpha_2 = r$$

has a rational detour solution with α_1 in an interval J , then

$$s(-\alpha_1) + (-\alpha_2) = -r$$

has a rational detour solution with $-\alpha_1$ in the reflected interval $-J$, because $\mathcal{S}$ is closed under negation. Reflecting back gives the original solution. Hence it suffices to treat

$$r = \frac{1 - k^4}{2k}.$$

The other sign follows by this reflection. The special value is

$$\tau = \frac{r}{s-1} = \frac{k^2 - 1}{2k}. \tag{7.9}$$

If $\tau \in I$, then put $\alpha_1 = \tau$. Then

$$\alpha_2 = r - s\tau \frac{1 - k^4}{2k} + k^2 \frac{k^2 - 1}{2k} \frac{1 - k^2}{2k} - \tau.$$

Since $\tau = (k - k^{-1})/2$, both τ and $-\tau$ lie in $\mathcal{S}$. Lemma 3.2 gives the desired point.

Assume now that $\tau \notin I$. Parametrize

$$\alpha_1 = \frac{t - t^{-1}}{2}.$$

The condition $\alpha_2 = r - s\alpha_1 \in \mathcal{S}$ gives the quartic

$$\begin{aligned} C_k : \quad W^2 &= k^6 t^4 + 2k^3(1 - k^4)t^3 \\ &\quad + (k^8 - 2k^6 - 2k^4 + 4k^2 + 1)t^2 \\ &\quad + 2k^3(k^4 - 1)t + k^6. \end{aligned} \tag{7.10}$$

The special point $\alpha_1 = \tau$ corresponds to

$$O = (t, W) = (k, k(k^2 + 1)).$$

Using O as the origin, (7.10) is birational to

$$\begin{aligned} E_k : \quad Y^2 - 2(k^4 - 2k^2 - 1)XY + 8k^6(k^4 - 1)Y \\ = X^3 - 4k^4(k^2 + 1)X^2. \end{aligned} \tag{7.11}$$

The explicit birational maps, nonsingularity of this model, and the torsion table used below are verified in Appendix A. The point

$$T = (0, 0)$$

has exact order 8. Explicitly, the first four multiples are

$$\begin{aligned} T &= (0, 0), \\ 2T &= (4k^4(k^2 + 1), -8k^4(k^2 + 1)^2), \\ 3T &= (4k^2(k^2 - 1)(k^2 + 1), -8k^4(k^2 - 1)(k^2 + 1)), \\ 4T &= (4k^6, -8k^8), \end{aligned}$$

and the negation formula on (7.11) is

$$-(X, Y) = (X, -Y + 2(k^4 - 2k^2 - 1)X - 8k^6(k^4 - 1)).$$

Thus

$$\begin{aligned} 5T &= (4k^2(k^2 - 1)(k^2 + 1), -8k^2(k^2 - 1)(k^2 + 1)^2), \\ 6T &= (4k^4(k^2 + 1), 0), \\ 7T &= (0, -8k^6(k^4 - 1)), \\ 8T &= \mathcal{O}. \end{aligned}$$

Since $4T \neq \mathcal{O}$, the order of T is exactly 8.

We now show that $E_k(\mathbb{Q})_{\text{tors}} = \langle T \rangle$. By Mazur's torsion theorem [4], an elliptic curve over $\mathbb{Q}$ with a rational point of order 8 can have torsion subgroup only

$$\mathbb{Z}/8\mathbb{Z} \quad \text{or} \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}.$$

Thus extra torsion would force extra rational 2-torsion. We solve the 2-torsion equation on (7.11) explicitly. A point (X, Y) has order 2 if and only if it equals its inverse, i.e.

$$2Y - 2(k^4 - 2k^2 - 1)X + 8k^6(k^4 - 1) = 0.$$

Substituting

$$Y = (k^4 - 2k^2 - 1)X - 4k^6(k^4 - 1)$$

into (7.11) gives

$$\begin{aligned} 0 = & -(X - 4k^6) \\ & \cdot (X^2 + (k^8 - 4k^6 - 2k^4 + 4k^2 + 1)X \\ & \quad - 4k^6(k^8 - 2k^4 + 1)). \end{aligned}$$

The root $X = 4k^6$ gives the known point $4T$. The quadratic factor has discriminant

$$(k - 1)^2(k + 1)^2(k^2 + 1)^4(k^4 + 6k^2 + 1).$$

Since $k \neq 0, \pm 1$, the quadratic gives additional rational 2-torsion if and only if

$$k^4 + 6k^2 + 1$$

is a square in $\mathbb{Q}$. Write $k = m/n$ in lowest terms. Then such a square would give integers m, n, z , with $mn \neq 0$, satisfying

$$m^4 + 6m^2n^2 + n^4 = z^2.$$

But then

$$(m + n)^4 + (m - n)^4 = 2z^2,$$

because

$$(m + n)^4 + (m - n)^4 = 2(m^4 + 6m^2n^2 + n^4).$$

The classical Fermat–Mordell quartic theorem says that

$$X^4 + Y^4 = 2Z^2$$

has only the trivial rational solutions $X = \pm Y$; see Mordell’s discussion [5] and Sidokhine’s exposition [9]. Applying this to $X = m + n$, $Y = m - n$ forces $m + n = \pm(m - n)$, hence $m = 0$ or $n = 0$, impossible. Hence there is no extra rational 2-torsion, and

$$E_k(\mathbb{Q})_{\text{tors}} = \langle T \rangle.$$

It remains to determine which α_1 -values can arise from these eight torsion points. The point at infinity $\mathcal{O}$ on E_k corresponds to the chosen origin $O = (k, k(k^2 + 1))$ on C_k , and hence to $t = k$. For all affine points with $Y \neq 0$, the inverse birational map has

$$t = k + \frac{2k(k^2 + 1)X}{Y}. \tag{7.12}$$

Substitution gives the following table:

torsion point	t -value	$\alpha_1 = (t - t^{-1})/2$
$\mathcal{O}$	k	τ
T	not in the affine t -chart	none
$2T$	0	undefined
$3T$	$-1/k$	τ
$4T$	$-1/k$	τ
$5T$	0	undefined
$6T$	not in the affine t -chart	none
$7T$	k	τ .

Here $t = 0$ is excluded because the parametrization $\alpha_1 = (t - t^{-1})/2$ requires $t \in \mathbb{Q}^\times$. Therefore the only finite α_1 -value represented by a rational torsion point is $\alpha_1 = \tau$. Therefore every finite rational point of C_k with $\alpha_1 \neq \tau$ is non-torsion.

Since $\tau \notin I$, the two endpoint values lie on the same side of τ , and neither is equal to τ . Hence both endpoint points are non-torsion. Lemma 7.1 gives infinitely many rational detour solutions with $\alpha_1 \in I$, and Lemma 3.2 gives the desired points P on UV . $\square$

Lemma 7.5 (Diagonal criterion for simple quadrilaterals). *Let Q be a simple non-degenerate quadrilateral with vertices*

$$P_1, P_2, P_3, P_4$$

in boundary order. Then Q is convex if and only if the two diagonals

$$P_1P_3, \quad P_2P_4$$

meet in their relative interiors.

Proof. If Q is convex, then Q is the convex hull of its four vertices. The segment P_1P_3 divides Q into the two triangles

$$\triangle P_1P_2P_3, \quad \triangle P_1P_3P_4,$$

which lie on opposite sides of the line P_1P_3 . Similarly, P_1 and P_3 lie on opposite sides of the line P_2P_4 . Therefore the two segments P_1P_3 and P_2P_4 cross, and because no three vertices are collinear, the crossing occurs in the relative interior of both segments.

Conversely, suppose Q is not convex. Since Q is simple and non-degenerate, its convex hull cannot have only two vertices, and it cannot have all four vertices as extreme points. Hence exactly one vertex of Q lies in the interior of the triangle formed by the other three vertices. After relabeling cyclically, assume this vertex is P_2 . Then P_2 lies in the interior of the triangle

$$\triangle P_1P_3P_4.$$

The diagonal P_1P_3 is one side of this triangle, while the other diagonal P_2P_4 is a segment from an interior point of the triangle to the opposite vertex P_4 . Since the triangle is convex, the segment P_2P_4 is contained in $\triangle P_1P_3P_4$; because P_2 is an interior point, every point of P_2P_4 except the endpoint P_4 lies in the interior of that triangle. Hence P_2P_4 is disjoint from the side P_1P_3 . Thus the diagonals do not meet in their relative interiors. The same argument applies no matter which vertex is the unique non-extreme vertex. Therefore, if the diagonals meet in their relative interiors, the quadrilateral must be convex. $\square$

7.4 The concave both-endpoints-torsion alternative

The phrase “both endpoint points are torsion” refers to the two rational points on the auxiliary genus-one curve lying above the two endpoint parameters of the interval I . This auxiliary condition can occur in the convex symmetric parallelogram with rational other diagonal, but it does not occur in the simple non-degenerate concave application.

Proposition 7.6 (The concave both-endpoints-torsion alternative is empty). *Let Q be a simple non-degenerate concave quadrilateral. In the normalized setup for the endpoint-line theorem, write*

$$U = (0, 0), \quad V = (D, 0), \quad A = (a, b), \quad B = (c, d), \quad b > 0 > d,$$

with UV the interior diagonal. In each exceptional endpoint-line case

$$r = 0, \quad s = -1, \quad r \neq 0, \quad 4r^2s = -(1 - s^2)^2,$$

the proof does not enter a finite both-endpoints-torsion alternative. In the torsion-controlled subcases, the two endpoint points on the corresponding auxiliary genus-one curve are not both torsion. Consequently these simple non-degenerate concave exceptional cases give infinitely many rational points on UV° , not finite examples.

Proof. Let

$$I = \left(-\frac{a}{b}, \frac{D-a}{b} \right)$$

be the interval of α -values corresponding to the open diagonal segment UV° . The special value

$$\tau = \frac{r}{s-1}$$

has a uniform geometric meaning in all exceptional endpoint-line cases:

$$x = a + b\tau$$

is the intersection point of the line AB with the x -axis, i.e. with the line containing the diagonal UV . For a simple non-degenerate concave quadrilateral with UV as the interior diagonal, this intersection point does not lie in the open diagonal segment. Indeed, the segment AB crosses the x -axis because $b > 0 > d$. If this crossing point also lay in UV° , then the two diagonals AB and UV would meet in their relative interiors, which is the convex alternative for a simple quadrilateral by Lemma 7.5. Therefore, in the simple non-degenerate concave case,

$$\tau \notin I.$$

We now check the three exceptional cases.

First suppose $r = 0$. Then

$$\tau = 0.$$

In the unequal-height subcase, the concordant-form torsion-value calculation says that, on each side of 0, there is at most one torsion value of the coordinate α . Since $0 \notin I$, the two distinct endpoint values

$$\lambda = -\frac{a}{b}, \quad \mu = \frac{D-a}{b}$$

lie on the same side of 0. They therefore cannot both be torsion. In the equal-height subcase $\rho = 1$, the proof is even simpler: the condition reduces to $\alpha \in \mathcal{S}$, and $\mathcal{S}$ is dense, so infinitely many points occur directly.

Next suppose $s = -1$ and $r \neq 0$. Put

$$T = -r.$$

The special value is

$$\tau = \frac{T}{2}.$$

The torsion-value calculation for

$$E_T : Y^2 = X^3 + (T^2 + 2)X^2 + X$$

says that, apart from the central value $T/2$, any additional finite torsion values occur as a symmetric pair

$$\eta, \quad T - \eta,$$

one on each side of $T/2$. If both endpoint values were torsion, then because $\tau \notin I$ they would have to lie on the same side of $\tau = T/2$. But the torsion table contains at most one torsion value on each side of $T/2$, a contradiction to $\lambda \neq \mu$. Equivalently, if the two endpoint values were the symmetric pair $\eta, T - \eta$, then $T/2$ would lie between them, forcing $\tau \in I$, which is the convex situation rather than the simple non-degenerate concave one.

Finally suppose

$$r \neq 0, \quad s \neq -1, \quad 4r^2s = -(1 - s^2)^2.$$

This is the quadratic exceptional identity. The Case III torsion table says that the only finite torsion value of the coordinate α is precisely

$$\alpha = \tau.$$

But $\tau \notin I$, while the endpoint values λ, μ lie in I . Hence neither endpoint point is torsion.

Thus, in every simple non-degenerate concave exceptional endpoint-line case, at least one endpoint point is non-torsion. The density lemma for a non-torsion endpoint then gives infinitely many rational points on the real component containing that endpoint. At the finite endpoint points used in these exceptional curves, the positive real detour branch is nonsingular and the coordinate α is a real analytic local parameter. Taking a sufficiently small one-sided neighborhood inside I therefore gives infinitely many rational detour values in I , hence infinitely many rational points on UV° at rational distance from all four vertices. $\square$

The finite both-endpoints-torsion phenomenon is real, but it belongs to the convex symmetric parallelogram with rational other diagonal, not to the simple non-degenerate concave cases. The example

$$U = (0, 0), \quad A = (18, 24), \quad V = (50, 0), \quad B = (32, -24)$$

in Theorem 6.8 is convex and has exactly one interior point on UV .

Theorem 7.7 (Exceptional endpoint theorem). *Assume the normalized endpoint-line setup and one of the exceptional conditions in (7.1). Then the open segment UV contains a rational point P such that $PA, PB \in \mathbb{Q}$. In the cases where the proof finds a non-torsion endpoint point, the segment contains infinitely many such rational points.*

Proof. If $r = 0$, this is Proposition 7.2. If $r \neq 0$ and $s = -1$, this is Proposition 7.3. Finally, if

$$r \neq 0, \quad s \neq -1, \quad 4r^2s = -(1 - s^2)^2,$$

this is Proposition 7.4. Since $s < 0$, these alternatives exhaust the exceptional cases. $\square$

Proof of the endpoint line theorem. By Lemma 2.1, reduce to the normalized setup. If the generic hypotheses hold, Proposition 4.2 gives the result. If not, since $s < 0$, one of the exceptional alternatives in (7.1) holds, and Theorem 7.7 gives the result. Transforming back by the inverse rational isometry preserves rationality of points and all distances. $\square$

8 Combining everything

Lemma 8.1 (Geometry of an interior diagonal). *Let Q be a simple non-degenerate quadrilateral, convex or concave, and let UV be an interior diagonal. If A and B are the other two vertices, then A and B lie on opposite sides of the line through UV .*

Proof. Let L be the line through U and V . Since Q is non-degenerate, neither A nor B lies on L . Suppose, for contradiction, that A and B lie on the same side of L . Let H be the closed half-plane bounded by L which contains A and B . The straight sides of Q are the four segments in one of the two cyclic orders

$$UA, AV, VB, BU \quad \text{or} \quad UB, BV, VA, AU.$$

Because a half-plane is convex, all four sides of Q lie in H . Therefore the boundary of Q is contained in H . The open half-plane opposite H is disjoint from the boundary and is connected and unbounded; by the Jordan curve theorem it is contained in the unbounded component of the complement of the boundary. Hence the bounded interior of the simple polygon lies in H . But if W is any point in the relative interior of UV , then every Euclidean disk centered at W contains points in the open half-plane opposite H , and those points are not in the interior of Q . Thus W cannot be an interior point of Q , contradicting the assumption that the relative interior of UV is contained in the interior of Q . Hence A and B lie on opposite sides of L . $\square$

Proof of Theorem 1.26. Let A and B be the two vertices of Q different from U, V . Because UV is an interior diagonal of a simple quadrilateral, the boundary order, up to reversal, is either

$$U, A, V, B \quad \text{or} \quad U, B, V, A.$$

Thus the four endpoint distances

$$UA, VA, UB, VB$$

are precisely the four side lengths of Q , and hence are integers. Also UV is an integer by hypothesis. By Lemma 8.1, the points A and B lie on opposite sides of the rational line through the integer points U, V . Since Q is non-degenerate, neither U nor V lies on the line AB .

We now separate the convex and concave cases.

First suppose that Q is concave. Theorem 3.1 applies directly to the rational line through U, V . Hence there is a rational point P on the open segment UV such that

$$PA, PB \in \mathbb{Q}.$$

Since $P \in UV$, and since U, V are rational points with $UV \in \mathbb{Q}$, the distances PU and PV are also rational. Indeed, write

$$P = (1 - \theta)U + \theta V.$$

Since P, U, V have rational coordinates and $U \neq V$, one coordinate of $V - U$ is nonzero, and that coordinate gives $\theta \in \mathbb{Q}$. Since $P \in UV^\circ$, we also have $0 < \theta < 1$. Hence

$$PU = \theta UV, \quad PV = (1 - \theta)UV$$

are rational. Because UV is an interior diagonal, P lies in the interior of Q . Thus the theorem is proved in the concave case.

Now suppose that Q is convex. Apply Lemma 2.1 to the diagonal UV . After, if necessary, reflecting in the x -axis and interchanging the names of A and B , we may assume

$$\begin{aligned} U &= (0, 0), & V &= (D, 0), & D &\in \mathbb{Q}_{>0}, \\ A &= (a, b), & B &= (c, d), & b &> 0 > d. \end{aligned}$$

The rational isometry preserves simplicity, convexity, and incidence, so in the normalized coordinates the boundary order may still be taken to be U, A, V, B , up to reversal. The four endpoint distances UA, VA, UB, VB remain rational. By convexity and Lemma 7.5, the diagonal AB meets the open segment UV . Put

$$s = \frac{b}{d}, \quad r = \frac{c-a}{d}.$$

Then $s < 0$.

If

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1-s^2)^2,$$

then Proposition 4.2 gives infinitely many rational points on the open segment UV whose distances from A and B are rational. For any one of them, the distances to U and V are rational because the point has rational x -coordinate and $0 < x < D$. Applying the inverse rational Euclidean isometry gives the required point in the original quadrilateral. Thus the convex non-exceptional case is complete. We may therefore assume from now on in the convex proof that the generic hypotheses fail.

It remains to consider the exceptional convex alternatives. Since $s < 0$, the reduction at the beginning of Section 5 shows that the only possibilities are

$$r = 0, \quad s = -1, \quad 4r^2s = -(1-s^2)^2.$$

If $r = 0$, Proposition 5.1 gives the required point. Assume next that $r \neq 0$. If

$$s \neq -1, \quad 4r^2s = -(1-s^2)^2,$$

then Proposition 5.5 gives the required point. Finally suppose $s = -1$. If $a + c \neq D$, then Proposition 5.7 gives the required point. If $a + c = D$, then Proposition 6.1 gives the required point.

In every convex subcase we have obtained a rational point on the normalized open segment UV whose distances from the four normalized vertices are rational. Applying the inverse rational Euclidean isometry sends it back to a rational point on the original open segment UV and preserves all four distances. Since UV is an interior diagonal, this point lies in the interior of Q . $\square$

Appendix

The appendix is purely computational and algebraic: it verifies the birational model and torsion table used only in Proposition 7.4.

A Verification of the Case III model and torsion table

This appendix verifies the birational model and the torsion table used in Proposition 7.4. Throughout $k \in \mathbb{Q}_{>0}$ and $k \neq 1$. In particular, $k \neq 0, \pm 1$, which is the nonvanishing condition used in the algebra below. Put

$$\begin{aligned} f_k(t) &= k^6 t^4 + 2k^3(1 - k^4)t^3 \\ &\quad + (k^8 - 2k^6 - 2k^4 + 4k^2 + 1)t^2 \\ &\quad + 2k^3(k^4 - 1)t + k^6. \end{aligned}$$

Thus

$$C_k : \quad W^2 = f_k(t)$$

is the quartic in (7.10). The chosen origin is

$$O = (k, k(k^2 + 1)).$$

We also write

$$u = t - k$$

and define

$$H_k(u) = k(k^2 + 1) + (-k^4 + 2k^2 + 1)u + k^3u^2. \quad (\text{A.1})$$

A direct expansion gives the identity

$$H_k(u)^2 = f_k(k + u) - 4k^5(k^2 - 1)u^3. \quad (\text{A.2})$$

The birational maps

Let E_k be the generalized Weierstrass curve

$$\begin{aligned} E_k : \quad Y^2 - 2(k^4 - 2k^2 - 1)XY + 8k^6(k^4 - 1)Y \\ = X^3 - 4k^4(k^2 + 1)X^2. \end{aligned} \quad (\text{A.3})$$

Equivalently, if

$$\mathcal{E}_k(X, Y) = Y^2 - 2(k^4 - 2k^2 - 1)XY + 8k^6(k^4 - 1)Y - X^3 + 4k^4(k^2 + 1)X^2,$$

then E_k is given by $\mathcal{E}_k(X, Y) = 0$. For this generalized Weierstrass equation the discriminant is

$$\Delta(E_k) 4096 k^{16} (k - 1)^2 (k + 1)^2 (k^2 + 1)^4 (k^4 + 6k^2 + 1). \quad (\text{A.4})$$

Since $k \in \mathbb{Q}_{>0}$ and $k \neq 1$, every factor in (A.4) is nonzero. Indeed, k , $k - 1$, and $k + 1$ are nonzero, while $k^2 + 1 > 0$ and $k^4 + 6k^2 + 1 > 0$. Thus E_k is a nonsingular elliptic curve over $\mathbb{Q}$.

Define a rational map

$$\Phi : C_k \dashrightarrow E_k$$

on the open set $u \neq 0$ by

$$X = \frac{2k(k^2 + 1)(W + H_k(u))}{u^2}, \quad (\text{A.5})$$

$$Y = \frac{4k^2(k^2 + 1)^2(W + H_k(u))}{u^3}. \quad (\text{A.6})$$

Substitution of (A.5) and (A.6) into the left side of (A.3) gives

$$\mathcal{E}_k(\Phi(t, W)) \frac{8k^3(k^2 + 1)^3(W + H_k(u))(f_k(k + u) - W^2)}{u^6}. \quad (\text{A.7})$$

Hence Φ maps C_k to E_k wherever it is defined.

Conversely, define

$$\Psi : E_k \dashrightarrow C_k$$

on the open set $Y \neq 0$ by

$$t = k + \frac{2k(k^2 + 1)X}{Y}, \quad (\text{A.8})$$

$$W = \frac{k(k^2 + 1)X^3 + 8k^7(k^2 - 1)(k^2 + 1)^2Y}{Y^2}. \quad (\text{A.9})$$

Substitution gives

$$W^2 - f_k(t) \frac{k^2(k^2 + 1)^2 G_k(X, Y) \mathcal{E}_k(X, Y)}{Y^4}, \quad (\text{A.10})$$

where

$$\begin{aligned} G_k(X, Y) = & -X^3 - 4k^4(k^2 + 1)X^2 + 2(k^4 - 2k^2 - 1)XY \\ & - Y^2 + 8k^6(k^4 - 1)Y. \end{aligned}$$

Therefore Ψ maps E_k to C_k wherever it is defined.

It remains to check that these maps are inverse on a common dense open set. Starting with a point of C_k , equations (A.5) and (A.6) give

$$\frac{2k(k^2 + 1)X}{Y} = u,$$

so (A.8) recovers $t = k + u$. Substituting (A.5) and (A.6) into (A.9), and using (A.2) together with $W^2 = f_k(k + u)$, recovers the original W . Thus $\Psi \circ \Phi$ is the identity where both maps are defined.

For the other composition, start with a point of E_k with $Y \neq 0$, and put

$$u = \frac{2k(k^2 + 1)X}{Y}.$$

Using (A.9), a direct simplification gives

$$\begin{aligned} W + H_k(u) = & \frac{k(k^2 + 1)}{Y^2} (X^3 + 4k^4(k^2 + 1)X^2 \\ & - 2(k^4 - 2k^2 - 1)XY + Y^2 + 8k^6(k^4 - 1)Y). \end{aligned}$$

On E_k , the expression in parentheses is $2X^3$, because $\mathcal{E}_k(X, Y) = 0$. Hence

$$W + H_k(u) = \frac{2k(k^2 + 1)X^3}{Y^2}. \quad (\text{A.11})$$

Substitution of (A.11) into (A.5) and (A.6) gives back the original (X, Y) . Therefore Φ and Ψ are inverse birational maps.

Finally, as $u \rightarrow 0$ on C_k along the branch through

$$O = (k, k(k^2 + 1)),$$

the formulas (A.5) and (A.6) have poles, so O maps to the point at infinity $\mathcal{O}$ of E_k . Conversely, the expansion of (A.8) and (A.9) at $\mathcal{O}$ gives $t \rightarrow k$ and $W \rightarrow k(k^2 + 1)$. Thus the chosen origin on C_k is indeed the point $\mathcal{O}$ on E_k .

The multiples of the distinguished torsion point

We now verify the torsion table on E_k . The curve (A.3) is in the generalized Weierstrass form

$$Y^2 + a_1XY + a_3Y = X^3 + a_2X^2,$$

with

$$a_1 = -2(k^4 - 2k^2 - 1), \quad a_3 = 8k^6(k^4 - 1), \quad a_2 = -4k^4(k^2 + 1).$$

The negation formula is therefore

$$-(X, Y) = (X, -Y + 2(k^4 - 2k^2 - 1)X - 8k^6(k^4 - 1)). \quad (\text{A.12})$$

The following points lie on E_k , by direct substitution:

$$\begin{aligned} P_1 &= (0, 0), \\ P_2 &= (4k^4(k^2 + 1), -8k^4(k^2 + 1)^2), \\ P_3 &= (4k^2(k^2 - 1)(k^2 + 1), -8k^4(k^2 - 1)(k^2 + 1)), \\ P_4 &= (4k^6, -8k^8), \\ P_5 &= (4k^2(k^2 - 1)(k^2 + 1), -8k^2(k^2 - 1)(k^2 + 1)^2), \\ P_6 &= (4k^4(k^2 + 1), 0), \\ P_7 &= (0, -8k^6(k^4 - 1)). \end{aligned} \quad (\text{A.13})$$

Equation (A.12) gives

$$-P_1 = P_7, \quad -P_2 = P_6, \quad -P_3 = P_5, \quad -P_4 = P_4.$$

Thus P_4 is a point of order 2.

We use the chord-and-tangent law. A line meets a generalized Weierstrass cubic in three points counted with multiplicity, and the sum of those three points is $\mathcal{O}$. The tangent at $P_1 = (0, 0)$ is $Y = 0$, and

$$\mathcal{E}_k(X, 0) = X^2(-X + 4k^4(k^2 + 1)).$$

Thus the tangent at P_1 meets the curve again at $P_6 = -P_2$, so

$$2P_1 = P_2.$$

The line through P_1 and P_2 is

$$Y = -2(k^2 + 1)X.$$

Substitution gives

$$\mathcal{E}_k(X, -2(k^2 + 1)X)X(-X + 4k^2(k^4 - 1))(-X + 4k^4(k^2 + 1)).$$

The third intersection point is $P_5 = -P_3$, so

$$P_1 + P_2 = P_3.$$

The line through P_1 and P_3 is

$$Y = -2k^2X.$$

Substitution gives

$$\mathcal{E}_k(X, -2k^2X) - X(-X + 4k^6)(-X + 4k^2(k^4 - 1)).$$

The third intersection point is $P_4 = -P_4$, so

$$P_1 + P_3 = P_4.$$

We have proved $2P_1 = P_2$, $3P_1 = P_3$, and $4P_1 = P_4$. Since P_4 is a nonzero point of order 2, it follows that $8P_1 = \mathcal{O}$. The negation relations above then give

$$5P_1 = -3P_1 = P_5, \quad 6P_1 = -2P_1 = P_6, \quad 7P_1 = -P_1 = P_7.$$

Thus

$$nP_1 = P_n \quad (1 \leq n \leq 7), \quad 8P_1 = \mathcal{O}.$$

Since $4P_1 = P_4 \neq \mathcal{O}$, the point $T = P_1$ has exact order 8.

The t -values of the torsion points

The inverse map Ψ gives

$$t = k + \frac{2k(k^2 + 1)X}{Y} \tag{A.14}$$

for affine points with $Y \neq 0$. The point $\mathcal{O}$ corresponds to the chosen origin O on C_k , hence to $t = k$. For the finite torsion points in (A.13), substitution in (A.14) gives

point on E_k	t -value	$\alpha_1 = (t - t^{-1})/2$
$\mathcal{O}$	k	$(k - k^{-1})/2$
$P_1 = T$	not in the $Y \neq 0$ chart	none
$P_2 = 2T$	0	undefined
$P_3 = 3T$	$-1/k$	$(k - k^{-1})/2$
$P_4 = 4T$	$-1/k$	$(k - k^{-1})/2$
$P_5 = 5T$	0	undefined
$P_6 = 6T$	not in the $Y \neq 0$ chart	none
$P_7 = 7T$	k	$(k - k^{-1})/2$

The entries $t = 0$ are excluded because the parametrization

$$\alpha_1 = \frac{t - t^{-1}}{2}$$

requires $t \in \mathbb{Q}^\times$. It remains to justify that the two torsion points with $Y = 0$, namely P_1 and P_6 , do not hide any additional finite t -values outside the chart (A.14). For a point in the domain of Φ , equations (A.5) and (A.6) give

$$\frac{Y}{X} = \frac{2k(k^2 + 1)}{u}$$

whenever $X \neq 0$. Therefore no finite point with $u \neq 0$ can map to P_6 , because P_6 has $X \neq 0$ and $Y = 0$. If a finite point with $u \neq 0$ mapped to $P_1 = (0, 0)$, then (A.5) would force

$$W + H_k(u) = 0.$$

Together with $W^2 = f_k(k + u)$, identity (A.2) would give

$$4k^5(k^2 - 1)u^3 = 0,$$

contradicting $u \neq 0$ and $k \neq 0, \pm 1$. Thus neither P_1 nor P_6 contributes a finite nonzero t -value with $u \neq 0$. If $u = 0$, then $t = k$, which gives the already-listed value $\alpha_1 = \tau$. Consequently the only finite α_1 -value produced by rational torsion is

$$\frac{k - k^{-1}}{2} = \frac{k^2 - 1}{2k} = \tau,$$

which is exactly the torsion-value assertion used in Proposition 7.4.

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